Space Exploration & Aerospace Engineering

Part V

Communication and Navigation Satellites

Satellites carry television, internet, phone calls and the timing signals that keep banks, power grids and phones synchronised. Part V explains communication payloads, broadband mega-constellations, direct-to-device service and the four global navigation systems: GPS, GLONASS, Galileo and BeiDou.



Chapters 17–18 · pages 302–342 of the complete manual

First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

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First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART V

Communication and Navigation Satellites

Satellites carry television, internet, phone calls and the timing signals that keep banks, power grids and phones synchronised. Part V explains communication payloads, broadband mega-constellations, direct-to-device service and the four global navigation systems: GPS, GLONASS, Galileo and BeiDou.

17 Communication Satellites

18 Global Navigation Satellite Systems

Image: NASA/KSC

17 CHAPTER Communication Satellites



Image: NASA/KSC (NASA/Kim Shiflett)

Communication satellites carry television to hundreds of millions of homes, connect ships, aircraft and remote communities to the internet, relay data from other spacecraft, and now connect ordinary smartphones directly from space. This chapter traces their evolution from passive balloons to geostationary "high-throughput" satellites delivering a terabit per second and to constellations of thousands of satellites in low orbit linked by lasers. It explains the engineering choices that shape every system — orbit, frequency band, propagation, link budget, payload architecture, frequency reuse and terminals — and the regulatory framework that shares the radio spectrum and orbits among nations. It closes with the current state of the industry in 2026: Starlink, OneWeb, Amazon Leo and the Chinese constellations, direct-to-device services, optical communications and data relay.

IN THIS CHAPTER

17.1 From Science Fiction to Global Infrastructure
17.2 Services and Markets
17.3 Choosing an Orbit
17.4 Frequencies, Propagation and Spectrum Sharing
17.5 Link Budgets and Terminals
17.6 The Communication Payload
17.7 LEO Broadband Constellations
17.8 Mobile Satellite Services and Direct-to-Device
17.9 Gateways, Teleports and Networks
17.10 Optical Communications and Data Relay

17.1 From Science Fiction to Global Infrastructure

In October 1945 the engineer and writer Arthur C. Clarke proposed in *Wireless World* that three satellites in a 24-hour orbit above the equator could relay radio signals around the whole world. Fifteen years later the idea began to come true. NASA's **Echo 1** (1960), a 30 m aluminised balloon, reflected radio signals passively between ground stations (Figure 17.1). AT&T's **Telstar 1** (1962)

was the first active communication satellite. It relayed the first live television pictures across the Atlantic. However, its low elliptical orbit meant it was visible from both continents for only minutes at a time. In 1963 Hughes' **Syncom 2** reached the first geosynchronous orbit, and in 1964 **Syncom 3** — in a true geostationary orbit — broadcast the Tokyo Olympic Games to the USA. In 1965 **Intelsat I** ("Early Bird") began commercial transatlantic telephone and television service with 240 voice circuits. The Soviet Union, whose territory lies at high latitudes, chose highly elliptical **Molniya** orbits for its own system from 1965 (Figure 17.3).



FIGURE 17.1 Inflation test of an Echo balloon satellite in 1958. Echo 1, launched in 1960, was a 30 m passive reflector — the first communication satellite experiment to relay voice between continents. Image: NASA/LRC (NACA).



FIGURE 17.2 A Syncom IV communication satellite after deployment from the Space Shuttle Discovery in 1984. The Syncom name dates back to the first geosynchronous satellites of 1963–64. Image: NASA/JSC.

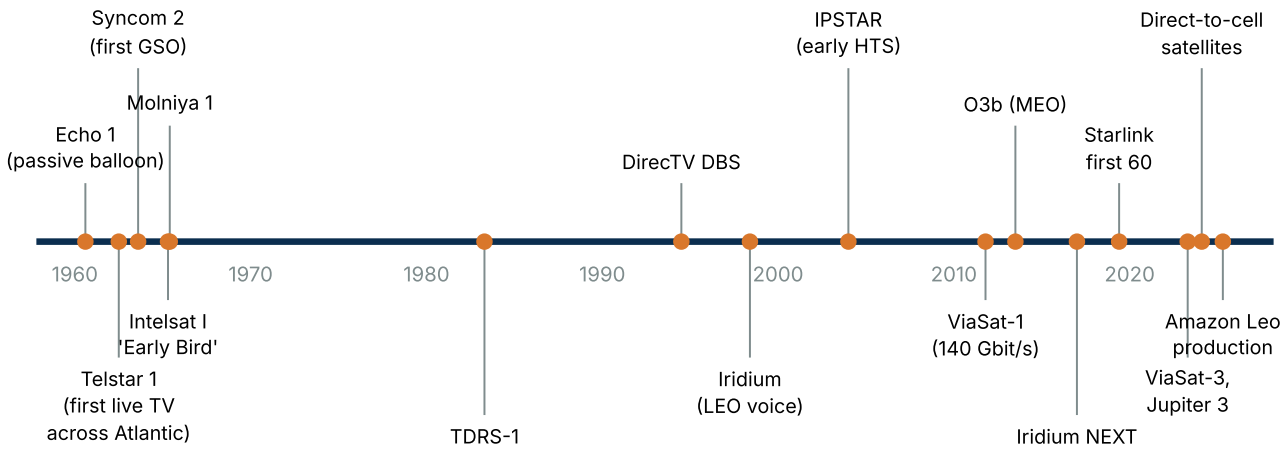


FIGURE 17.3 Milestones in satellite communications.

For half a century, communication satellites were dominated by a few hundred large geostationary satellites operated by international organisations (Intelsat, Inmarsat, Eutelsat — later privatised) and national and commercial operators. They carried international telephone trunks (until undersea fibre took over from the 1990s), distributed television to cable head-ends and directly to homes, and provided mobile communications to ships and aircraft. In the 2010s two revolutions transformed the industry: **high-throughput satellites (HTS)** in GEO, which multiplied capacity a hundredfold using many small spot beams; and **low-Earth-orbit broadband constellations** of hundreds to thousands of mass-produced satellites. In 2026 satellite communications remain the largest commercial space market, generating more than $100 billion a year in services and equipment.



FIGURE 17.4 The Space Shuttle crew of STS-49 captured the stranded Intelsat VI satellite in 1992 during the first three-person spacewalk, attached a new perigee motor and sent it on to geostationary orbit. Communication satellites were among the Shuttle's main commercial payloads. Image: NASA/MSFC.

17.2 Services and Markets

Satellite communications serve several distinct markets, each defined in the ITU Radio Regulations as a "service" with its own frequency allocations (Table 17.1).

TABLE 17.1 Satellite communication services.

Service	Typical users	Main bands	Example systems
Fixed Satellite Service (FSS): trunking, backhaul, enterprise networks, video contribution	Telecom operators, broadcasters, businesses	C, Ku, Ka	SES, Intelsat, Eutelsat, APT, Telesat
Broadcasting Satellite Service (BSS): direct-to-home TV and radio	Households	Ku (TV), S/L (radio)	SES Astra, DirecTV, Dish, Sky, SiriusXM
Satellite broadband	Homes, businesses, communities, governments	Ku, Ka, Q/V, E	Starlink, OneWeb, Amazon Leo, Viasat, Hughes
Mobility: maritime and aeronautical connectivity	Ships, oil rigs, airlines, business jets	L, Ku, Ka	Inmarsat (Viasat), Panasonic, Starlink, OneWeb

Service	Typical users	Main bands	Example systems
Mobile Satellite Service (MSS): handheld voice and data, IoT	Remote workers, emergency services, militaries	L, S	Iridium, Globalstar, Inmarsat, Thuraya
Direct-to-device (D2D)	Ordinary smartphones	Terrestrial cellular bands, S, L	Starlink Direct to Cell, AST SpaceMobile, Lynk, Apple–Globalstar
Government and military satellite communications	Armed forces, civil protection	UHF, X, Ka, EHF	WGS, AEHF/ESS, Skynet, Syracuse (Chapter 19)
Data relay	Other spacecraft (ISS, EO satellites)	S, Ku, Ka, optical	TDRS, EDRS, Tianlian, Luch

Broadcasting was long the most profitable segment, but linear satellite TV is declining as viewers move to internet streaming, while satellite broadband — especially LEO broadband — is growing rapidly. Starlink alone reported about 12 million customers in more than 160 countries by June 2026.

17.3 Choosing an Orbit

The orbit determines coverage, latency, the number of satellites needed and the cost of terminals (Chapter 11).

- **Geostationary orbit (GEO)** allows a satellite to cover about a third of the Earth continuously, and simple fixed dishes to point at it. Three satellites cover the world except the polar regions. The penalty is distance: radio signals take about a quarter of a second to travel up and down. So a round trip through a GEO satellite (user → satellite → gateway → satellite → user) takes at least 477 ms, plus processing. This delay is noticeable in conversations and online gaming. The large distance also requires big antennas and high power.
- **Medium Earth orbit (MEO)** offers a compromise. SES's **O3b** and **O3b mPOWER** satellites, in an 8 062 km equatorial orbit, provide fibre-like links with round-trip latency of about 110–150 ms to cruise ships, mobile operators and governments.
- **Low Earth orbit (LEO)** at 500–1 200 km gives latencies comparable with terrestrial networks (Figure 17.5) and allows smaller terminals. But each satellite is visible from a given place for only a few minutes. So hundreds to thousands of satellites, steerable user antennas and handovers between satellites are required.
- **Highly elliptical orbits (HEO)** such as Molniya and Tundra serve high latitudes. Space Norway's two Arctic Satellite Broadband Mission satellites (2024), carrying military and commercial payloads, use such orbits to provide continuous coverage of the Arctic.

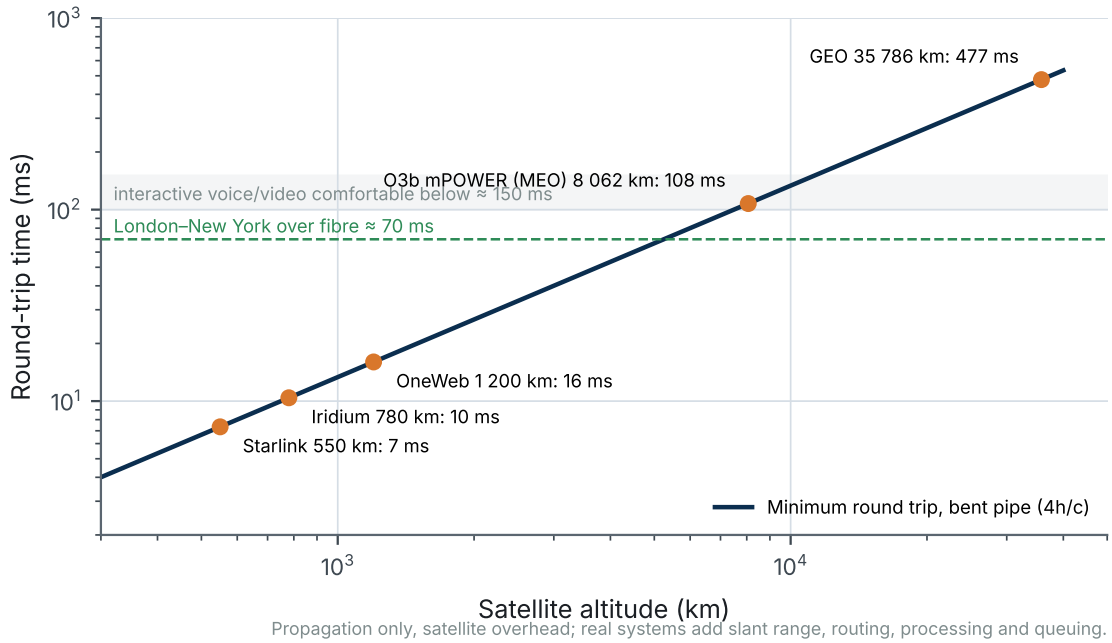


FIGURE 17.5 Minimum round-trip propagation time through a bent-pipe satellite versus altitude. LEO systems approach the latency of terrestrial fibre.

17.4 Frequencies, Propagation and Spectrum Sharing

17.4.1 Frequency Bands

Satellite communications use bands from VHF to millimetre waves and, increasingly, optical wavelengths (Chapter 7). Lower bands (L, S) propagate well, penetrate foliage and work with small, non-directional antennas — ideal for mobile services — but offer little bandwidth. Higher bands (Ku, Ka, Q/V, E) offer gigahertz of bandwidth and allow small, high-gain antennas, but suffer from rain. The industry has migrated steadily upwards: C-band for early trunking, Ku-band for television from the 1980s, Ka-band for broadband from the 2000s, and Q/V- and E-band for gateway links in the newest systems.

17.4.2 Rain Fade

Above about 10 GHz, raindrops absorb and scatter radio waves. The **specific attenuation** (dB/km) grows rapidly with frequency and rain rate (Figure 17.6). A heavy tropical downpour of 100 mm/h can attenuate a Ka-band signal by 30–50 dB — enough to break any link. The same rain costs only a few dB at Ku-band and almost nothing at C-band. Because heavy rain is rare and localised, systems are designed for a specified **availability** (for example 99.7 % of the average year, about 26 hours of outage). They use statistical rain models from ITU-R Recommendations, and they combat fades with:

- **adaptive coding and modulation (ACM)**, which switches to more robust waveforms when the link degrades (standardised in DVB-S2 and DVB-S2X);
- **uplink power control**, which raises the transmitted power during a fade;
- **site diversity**, using two gateways several tens of kilometres apart, since intense rain cells rarely cover both.

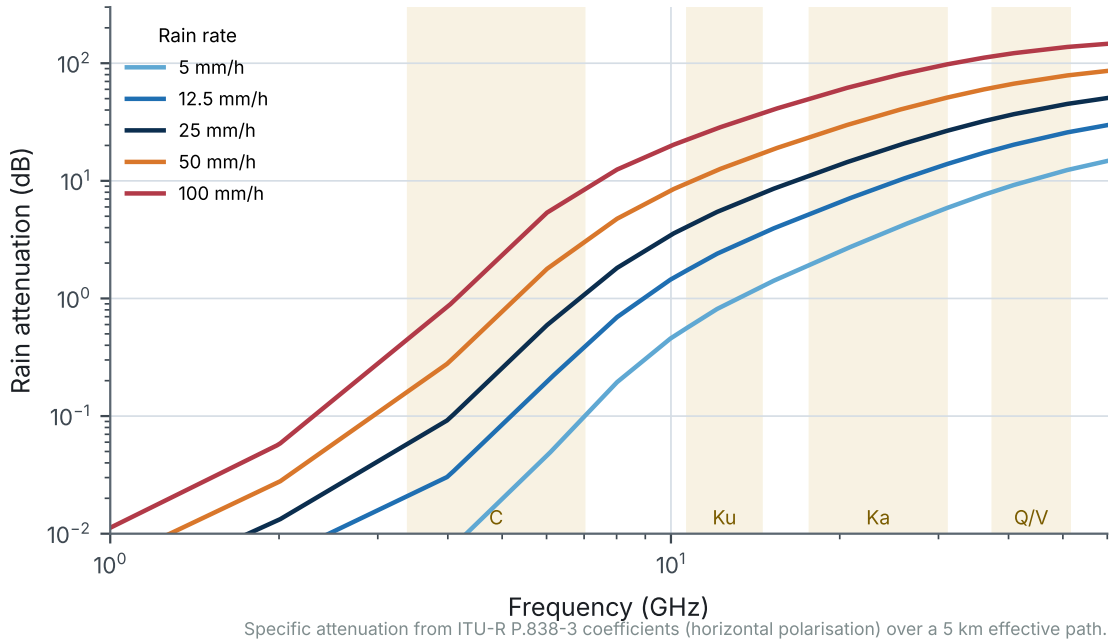


FIGURE 17.6 Rain attenuation versus frequency for different rain rates. Attenuation increases dramatically above about 10 GHz.

17.4.3 Sharing the Spectrum and the Orbits

Radio spectrum and orbital positions are limited natural resources shared by all nations. The **International Telecommunication Union (ITU)**, a United Nations agency, maintains the **Radio Regulations** — an international treaty that allocates frequency bands to services — and the **Master International Frequency Register**. A satellite operator, through its national administration, must file its planned network with the ITU, **coordinate** it with earlier networks that might suffer interference, and bring it into use within seven years. Geostationary positions are typically spaced 2–3° apart, and coordination between neighbouring operators is a complex negotiation. Every three to four years a **World Radiocommunication Conference** (WRC-23 in Dubai; WRC-27 next) revises the Regulations — for example to allocate spectrum for direct-to-device services or to set rules for lunar communications.

Non-geostationary constellations must protect GEO networks. ITU **equivalent power-flux-density (EPFD)** limits cap the interference that their combined satellites may cause to GEO terminals. In practice, this means that a LEO system must not transmit along the line of sight between a ground terminal and the geostationary arc (Figure 17.7). Large constellations must also meet **milestones**. Since 2019, ITU rules require 10 % of a constellation to be deployed within two years of the end of the seven-year bring-into-use period. They also require 50 % within five years and 100 % within seven years. National licences add their own conditions. In the USA, for example, the FCC required Amazon's constellation to have half of its satellites (about 1 600) in orbit by July 2026. With roughly 300–400 in orbit, Amazon obtained a conditional waiver in 2026.

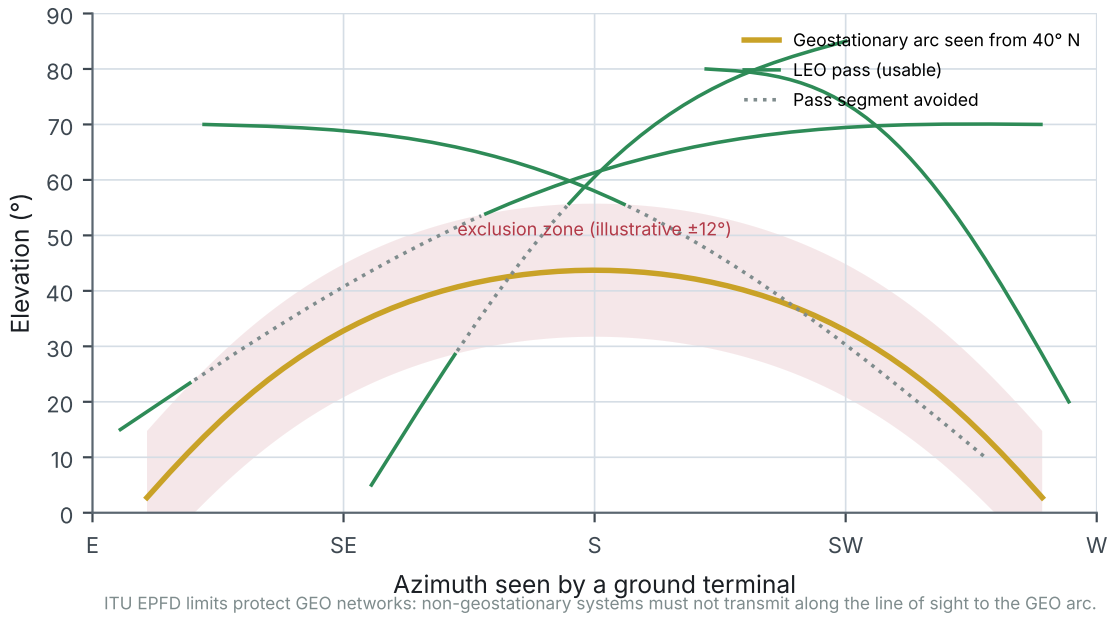


FIGURE 17.7 A ground terminal's view of the sky. To protect geostationary networks, LEO systems avoid using satellites that appear near the geostationary arc (illustrative exclusion zone).

17.5 Link Budgets and Terminals

The link-budget method of Chapter 7 determines whether a link closes. Figure 17.8 shows the budget of a typical Ku-band direct-to-home television downlink as a "waterfall". A satellite EIRP of 52 dBW is attenuated by a free-space loss of about 206 dB over 38 000 km. A 60 cm dish with a low-noise block converter recovers about 36 dB of gain against a system noise temperature of about 120 K. In a 27 MHz channel, the resulting carrier-to-noise ratio is about 15 dB. This is enough for the DVB-S2 8PSK rate-3/4 waveform, with a margin of more than 6 dB for rain.

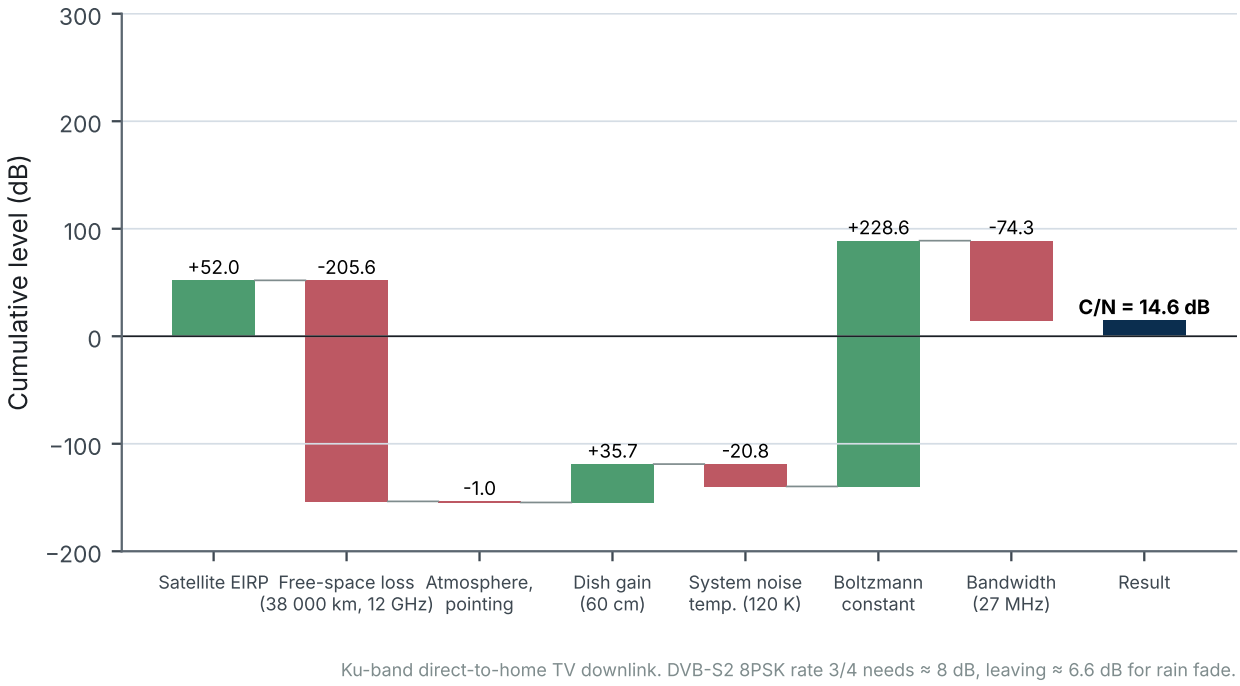


FIGURE 17.8 Link budget of a Ku-band direct-to-home television downlink, shown as a waterfall of gains and losses.

Modern systems adapt their waveforms continuously. The **DVB-S2X** standard offers more than 100 combinations of modulation (from QPSK to 256APSK) and coding, achieving spectral efficiencies from below 0.5 to more than 5 bit/s/Hz, within about 1 dB of the Shannon limit. Since 2022 the 3GPP mobile standards (Release 17 onward) have included **non-terrestrial networks (NTN)**, allowing satellites to be integrated into 5G — the basis for direct-to-device services.

17.5.1 User Terminals

Terminals have changed as much as satellites (Figure 17.9). The 60 cm dish for satellite TV, fixed to a wall and pointed at a GEO satellite, costs a few tens of dollars. A LEO broadband system needs an antenna that follows satellites across the sky and switches between them every few minutes. The breakthrough was the low-cost **electronically steered phased array**. Starlink's flat terminal contains about a thousand or more antenna elements. Their signals are phase-shifted by custom chips to form and steer beams with no moving parts. Through mass production, the terminal's cost fell from thousands of dollars to a few hundred. Aircraft and ships use larger flat-panel or mechanically steered terminals.

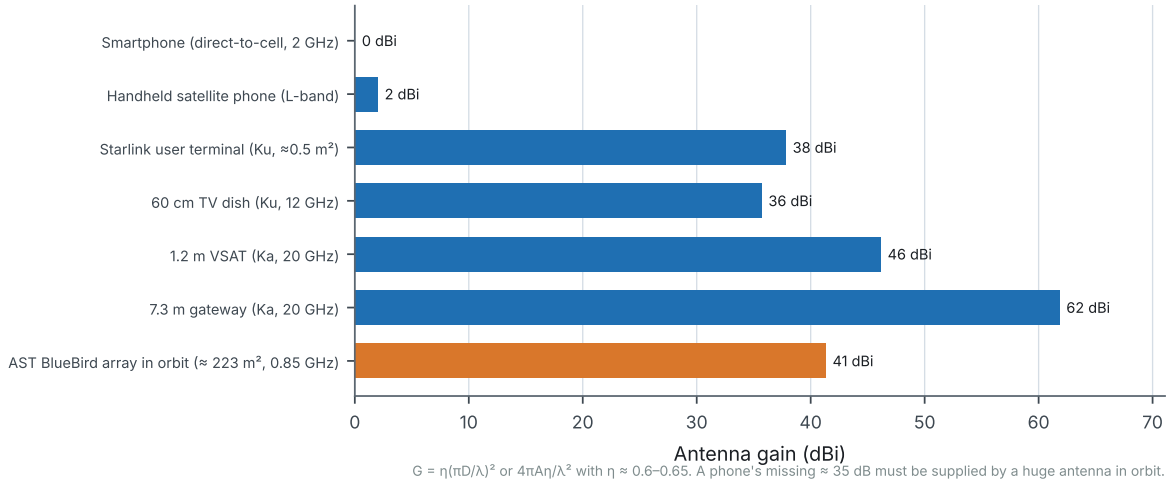


FIGURE 17.9 Antenna gain of typical terminals. Direct-to-device service must close the link with a smartphone's near-isotropic antenna, so the satellite antenna must be enormous.

17.6 The Communication Payload

17.6.1 Transponders: Bent Pipe and Regenerative

The heart of a classic communication satellite is the **transponder** (Figure 17.10). In a **bent-pipe** (transparent) design, the receive antenna collects the uplink signals and a low-noise amplifier boosts them. Filters and frequency converters then shift them from the uplink to the downlink band (for example from 14 GHz to 12 GHz at Ku-band). An **input multiplexer** divides the band into channels, typically 36–72 MHz wide. Each channel is amplified by a **travelling-wave-tube amplifier (TWTA)** or a solid-state amplifier. Finally, an **output multiplexer** recombines the channels for the transmit antenna. A bent pipe does not care what signals it carries, which makes it flexible and long-lived — a satellite launched for analogue television can carry digital broadband 15 years later.

A **regenerative** payload demodulates and decodes the uplink signals, switches or routes the data, and re-modulates it for the downlink. Regeneration improves link quality (uplink and downlink noise no longer add), enables on-board routing between beams and inter-satellite links, and is essential for networks such as Iridium and Starlink. **Digital transparent processors** sit between the two: they digitise the band and route and filter channels flexibly without decoding them.

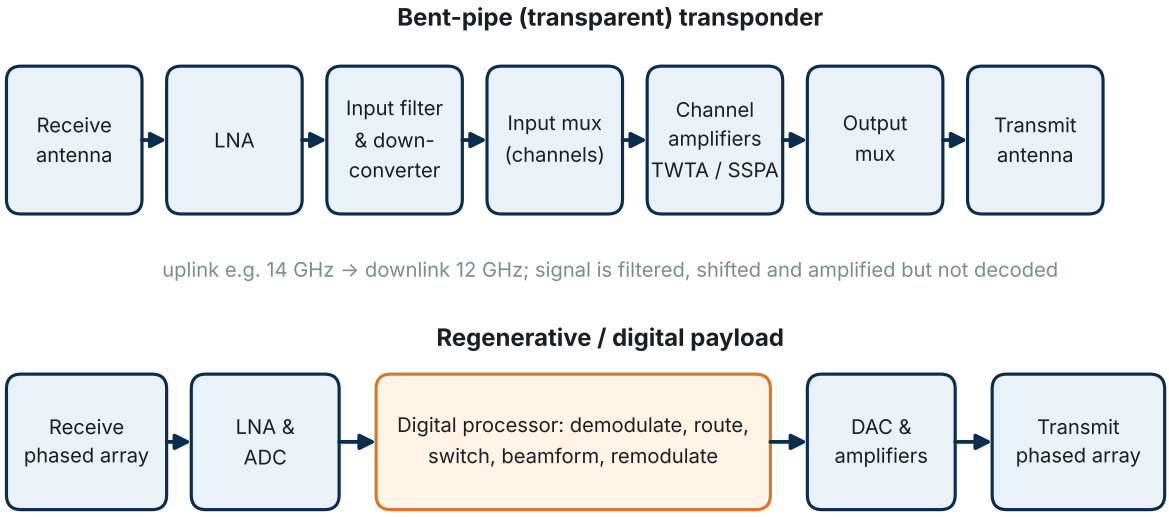


FIGURE 17.10 Architecture of bent-pipe and regenerative communication payloads.

TWTAs remain the workhorse high-power amplifiers in space because of their efficiency (60–70 %) and power (tens to hundreds of watts per tube). Gallium-nitride solid-state amplifiers are increasingly used, especially in phased arrays where many small amplifiers are distributed over the antenna.

17.6.2 Antennas, Spot Beams and Frequency Reuse

Early satellites used a single **global beam** or a few regional beams. Capacity is limited by bandwidth, so the key to more capacity is to **reuse** the same frequencies many times. **Multi-beam antennas** — large reflectors fed by clusters of horns, or phased arrays — form tens to hundreds of narrow **spot beams**, each a few hundred kilometres wide. Adjacent beams use different frequency sub-bands or polarisations; beams that are far enough apart reuse the same ones. In the classic **four-colour** scheme (Figure 17.11), the available spectrum is split into two sub-bands and two polarisations. So each beam uses half the spectrum, and the whole band is reused about a quarter as many times as there are beams. A satellite with 100 beams thus reuses its spectrum about 25 times. Smaller beams also have higher antenna gain, so users can have smaller terminals and higher data rates.

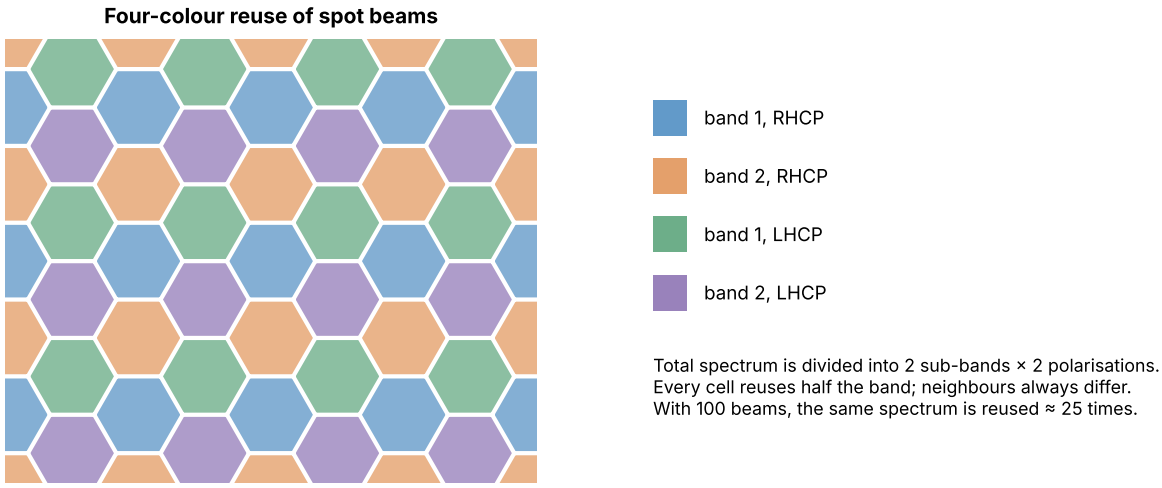


FIGURE 17.11 Four-colour frequency reuse with spot beams. Neighbouring cells always use different sub-band and polarisation combinations.

17.6.3 High-Throughput Satellites

Applying this principle at Ka-band produced **high-throughput satellites (HTS)**. Eutelsat's KA-SAT (2010) had 82 spot beams and 90 Gbit/s of capacity; Viasat's ViaSat-1 (2011) offered 140 Gbit/s — more than all other satellites over North America combined at the time. Capacity per satellite has grown roughly a hundred-thousand-fold since Early Bird (Figure 17.12). Hughes' Jupiter 3 (EchoStar XXIV, 2023) provides about 500 Gbit/s. Viasat's **ViaSat-3** class was designed for more than 1 Tbit/s per satellite. But the first satellite's huge deployable reflector failed to deploy fully in 2023, sharply reducing its capacity. This is a reminder of the risks of large deployable structures (Chapter 5). The second ViaSat-3 was launched in late 2025.

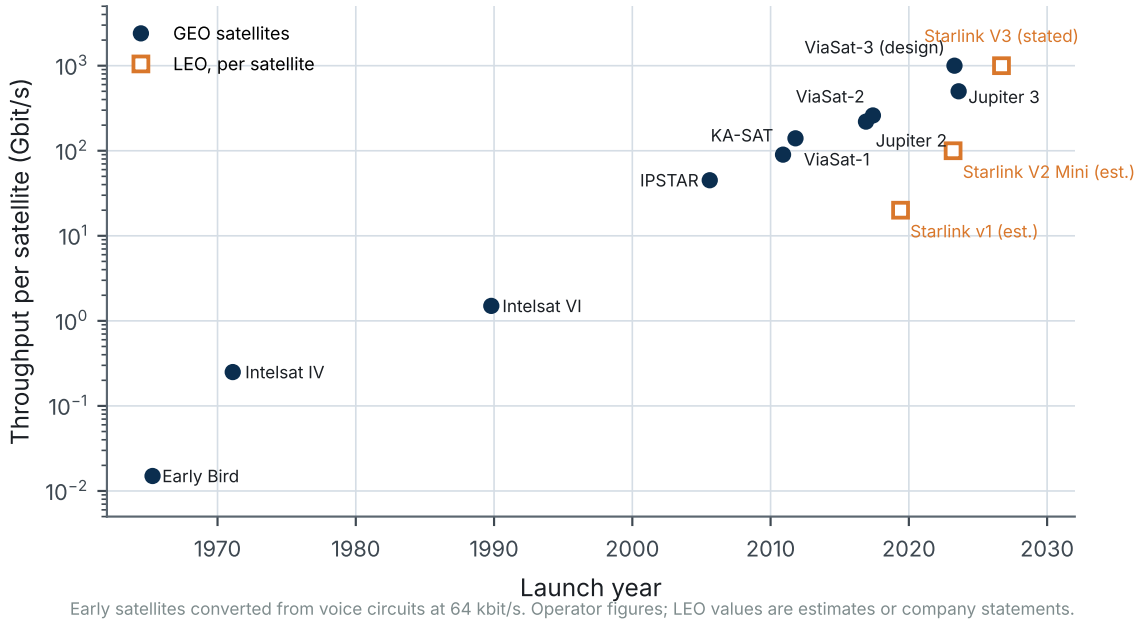


FIGURE 17.12 Throughput per satellite from Early Bird (1965) to modern HTS in GEO and to individual LEO broadband satellites.

17.6.4 Software-Defined Satellites

A GEO satellite must serve markets for 15 years that cannot be predicted when it is ordered. **Software-defined** or **flexible** satellites include Eutelsat Quantum (2021), Airbus OneSat, Thales Alenia Space's Space Inspire and Boeing's 702X. They use digital processors and active phased-array antennas, so that coverage, beam shapes, frequencies and power can be reconfigured in orbit. This lets them follow demand, or even follow a moving ship or aircraft. Standardised designs also shorten manufacturing times from about three years to about 18 months. At the other end of the scale, start-ups such as **Astranis** build small (about 400 kg) "MicroGEO" satellites that serve a single customer or country with a dedicated software-defined payload.

17.6.5 The Geostationary Industry in Transition

The rise of LEO constellations has put pressure on traditional GEO operators. Orders for large commercial GEO satellites fell from 20–25 per year in the 2000s and early 2010s to around ten or fewer in the 2020s. Operators have also consolidated: Eutelsat merged with OneWeb (2023), Viasat acquired Inmarsat (2023), and SES completed its acquisition of Intelsat in July 2025. Most GEO operators now offer "multi-orbit" services combining their GEO capacity with LEO or MEO partners.

17.7 LEO Broadband Constellations

17.7.1 Earlier Attempts

LEO communication constellations are not new. Iridium (66 satellites, 1998), Globalstar (48 satellites, 1999) and Orbcomm (1990s) were built for mobile telephony and messaging, and Teledesic planned a constellation of 840 broadband satellites in the 1990s. Iridium and Globalstar went bankrupt shortly after entering service — undercut by cheaper terrestrial mobile networks — although both later recovered and replaced their constellations. What changed in the 2010s was the cost of building and launching satellites, the availability of cheap phased arrays, and soaring demand for broadband.

17.7.2 Starlink

SpaceX's **Starlink** has become by far the largest satellite system ever built (Figure 17.13, Figure 17.14). Its first 60 production-type satellites were launched in May 2019; by mid-October 2026 about 11 000 were in orbit, launched mostly by SpaceX's own reusable Falcon 9 rockets in batches of 20–30 satellites. Key features of its design include:

- **Flat-panel satellites** stacked in the fairing and released together. Each has phased-array antennas for user beams (Ku-band), parabolic or phased-array antennas for gateways (Ka- and E-band), a single large solar array and **Hall-effect thrusters**. The thrusters use krypton or argon for orbit raising, station-keeping and de-orbiting.
- **Low altitudes** — mostly 480–560 km, in "shells" at inclinations of 43°, 53°, 70° and 97.6° (Figure 17.15). At these altitudes, failed satellites re-enter within about five years by atmospheric drag, which limits long-term debris; in 2026 SpaceX began lowering many satellites towards about 480 km.

- **Laser inter-satellite links** on all recent satellites, forming an optical mesh network that carries traffic across oceans and to places without nearby gateways (Figure 17.16).
- **Rapid iteration**: successive versions (v0.9, v1.0, v1.5, V2 Mini) have increased capacity per satellite. The much larger **V3** satellites, designed for about 1 Tbit/s of downlink capacity each, require the Starship launch vehicle; the first V3 satellites were deployed on Starship's fourteenth flight in September 2026.
- **Direct to Cell** satellites with large antennas that connect to ordinary phones (see below).

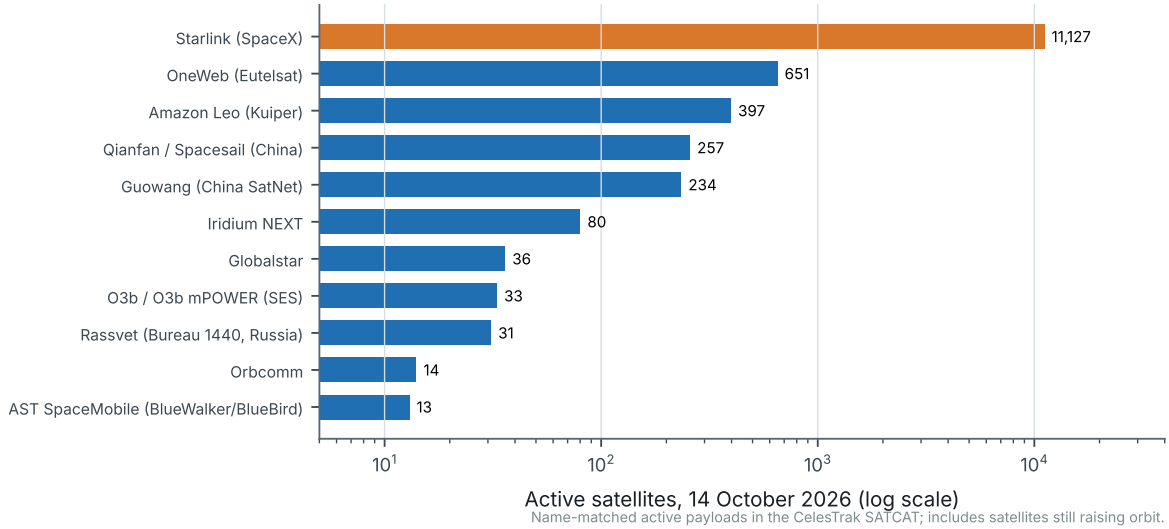


FIGURE 17.13 Active satellites in the main broadband and mobile communication constellations in October 2026 (logarithmic scale).

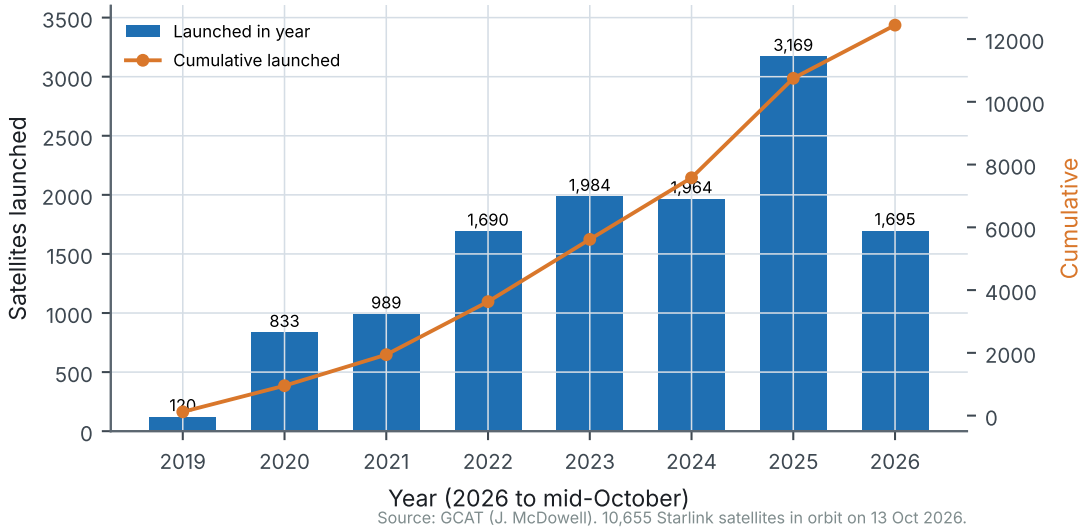
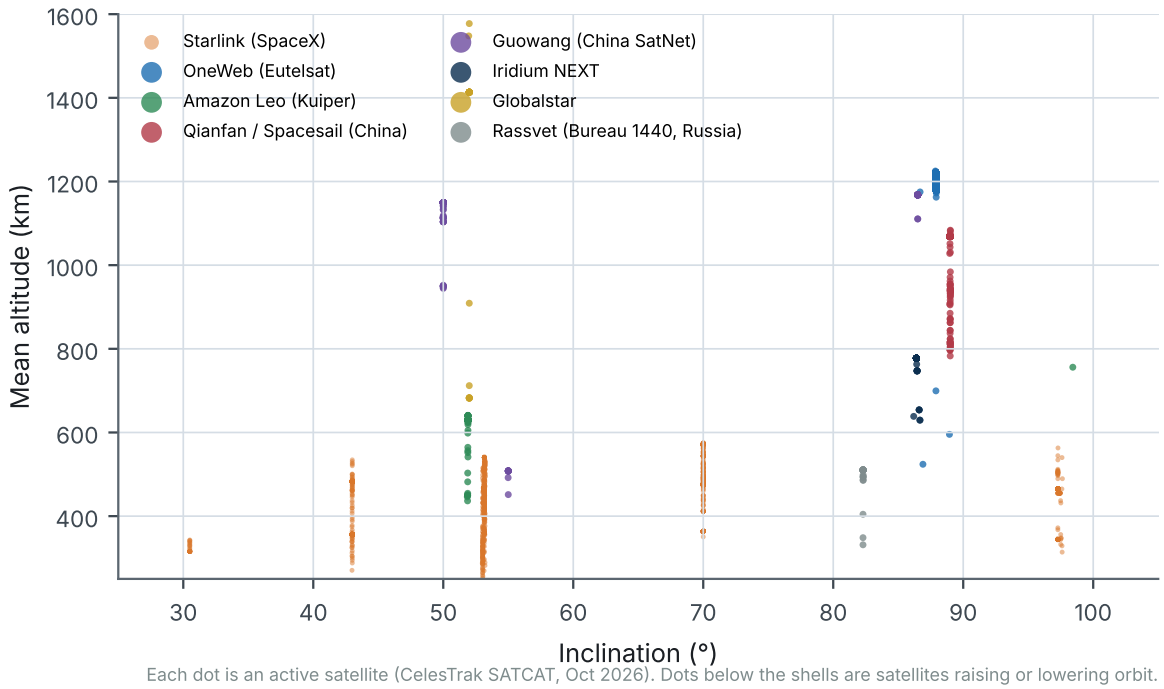


FIGURE 17.14 Starlink satellites launched per year and cumulatively, 2019 to mid-October 2026.



Each dot is an active satellite (CelesTrak SATCAT, Oct 2026). Dots below the shells are satellites raising or lowering orbit.

FIGURE 17.15 Orbital shells of the main LEO communication constellations, plotted from the October 2026 satellite catalogue. Each constellation uses a few distinct combinations of altitude and inclination.

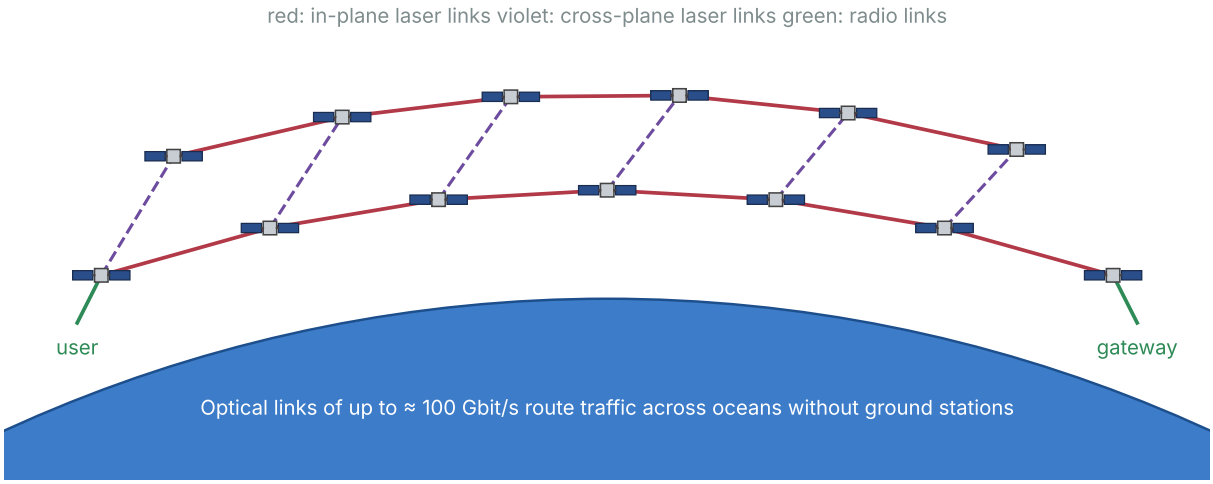


FIGURE 17.16 Laser inter-satellite links create an optical mesh network in space, routing traffic between distant users and gateways.

17.7.3 Other Constellations

- **OneWeb** (Eutelsat): about 650 satellites at 1 200 km in near-polar orbits (87.9°), built with Airbus. Rather than selling directly to consumers, OneWeb serves telecom operators, governments, enterprises and mobility providers. A second generation is being procured.
- **Amazon Leo** (formerly Project Kuiper): authorised for 3 236 satellites in shells at 590–630 km. Amazon began full-scale launches in April 2025, using Atlas V, Falcon 9, Vulcan and other rockets, and renamed the system in November 2025. About 400 satellites were in orbit by October 2026; launch-vehicle shortages, including the loss of a New Glenn rocket in a pad explosion in May 2026, slowed deployment.

- **China's Guowang** (China Satellite Network Group, about 13 000 satellites planned) and **Qianfan** (Spacesail, about 14 000 planned) began deploying in 2024; each had more than 200 satellites in orbit by October 2026.
- **Telesat Lightspeed** (Canada): about 200 satellites built by MDA in polar and inclined orbits at about 1 000–1 300 km, serving enterprise and government customers, with launches from 2026.
- **IRIS²** (EU): a multi-orbit secure-connectivity constellation of roughly 300 satellites in LEO and MEO for governmental and commercial services, with first launches expected around 2029–2030.
- **Rassvet** (Bureau 1440, Russia) began launching in 2025.

TABLE 17.2 Major non-geostationary communication constellations (approximate).

System	Operator	Satellites (planned / active Oct 2026)	Altitude	Frequency bands	Market
Starlink	SpaceX (USA)	≈ 12 000 authorised (Gen 1), up to ≈ 30 000 Gen 2 proposed / ≈ 11 000	340–570 km	Ku, Ka, E; PCS/AWS for D2C	Consumer, enterprise, mobility, government, D2C
OneWeb	Eutelsat (France/UK)	648 Gen 1 / ≈ 650	1 200 km	Ku, Ka	Wholesale, enterprise, government
Amazon Leo	Amazon (USA)	3 236 / ≈ 400	590–630 km	Ka	Consumer, enterprise, government
Guowang	China SatNet	≈ 13 000 / ≈ 230	500–1 150 km	Ku, Ka, Q/V	National broadband
Qianfan (Spacesail)	Shanghai SSST	≈ 14 000 / ≈ 260	≈ 800–1 100 km	Ku, Ka	National and international broadband
Lightspeed	Telesat (Canada)	198 / first launches 2026	≈ 1 000–1 300 km	Ka	Enterprise, government
Iridium NEXT	Iridium (USA)	66 + spares / ≈ 80	780 km	L, Ka	Mobile voice, IoT, safety
O3b mPOWER	SES (Luxembourg)	13 / 13	8 062 km (MEO)	Ka	Enterprise, cruise, government
IRIS²	EU	≈ 300 (LEO + MEO) / —	LEO and MEO	Ka and others	Government and commercial

17.7.4 Challenges

Mega-constellations raise new problems. Their satellites are visible in astronomical images as streaks, and their radio emissions can interfere with radio telescopes. Operators have darkened satellites, adjusted orientations and coordinated with observatories. Even so, the effect on wide-field astronomy is significant (Chapter 38). The sheer number of satellites makes **collision avoidance** a continuous activity. Starlink satellites perform tens of thousands of automated avoidance manoeuvres per year. The number of satellites also raises concerns about the long-

term sustainability of LEO (Chapter 21). The re-entry of thousands of satellites per year is also adding aluminium oxide to the upper atmosphere, a possible environmental effect now being studied.

17.7.5 The Economics of a Constellation

A LEO broadband system is a capital-intensive bet. Building, launching and operating several thousand satellites, plus gateways and user terminals, costs on the order of $10–30 billion before the system covers its costs, and satellites must be replaced every five to seven years. Three factors dominate the economics:

1. **Cost per satellite and per launch.** Starlink, OneWeb and Amazon all built factories producing several satellites per day on assembly lines. This mass production has reduced satellite costs to a few hundred thousand to a few million dollars each. Reusable launchers have also cut launch costs per kilogram by a factor of five to ten (Chapter 34).
2. **Capacity per satellite.** Because most of the fleet flies over oceans and empty land at any moment, revenue comes only from the fraction over paying customers. Larger satellites with more capacity improve this ratio dramatically, which is why SpaceX's V3 satellites depend on Starship.
3. **Terminal cost.** A phased-array terminal that costs more to make than the customer pays for it must be subsidised. Driving terminal costs down from thousands to a few hundred dollars was essential to reaching consumers.

Vertical integration — one company building the satellites, the rockets, the terminals and the network — gave SpaceX a decisive cost advantage. Governments increasingly view sovereign broadband constellations (China's Guowang and Qianfan, the EU's IRIS^2) as strategic infrastructure, not just commercial ventures.

17.7.6 Satellite IoT and Machine-to-Machine Services

A separate class of constellations serves the **Internet of Things (IoT)**: tracking containers, livestock, wildlife, pipelines, ships and environmental sensors in remote areas with very small, low-power devices. Systems include Orbcomm, Iridium's short-burst data, Kinéis (France, 25 nanosatellites launched in 2024, continuing the Argos system used for wildlife tracking since 1978), Astrocast, Myriota and Swarm (acquired by SpaceX). Some use standard LoRa or NB-IoT protocols so that cheap terrestrial chips can talk to satellites. Data rates are tiny — bytes to kilobytes per day — but the number of devices could reach tens of millions.

17.8 Mobile Satellite Services and Direct-to-Device

17.8.1 Classic Mobile Satellite Systems

Mobile satellite services use L- and S-band frequencies, which suit small, non-directional antennas. **Inmarsat** (founded in 1979 for maritime safety, now part of Viasat) provides voice, data and safety services to ships and aircraft from GEO, including the Global Maritime Distress and Safety System. **Iridium**'s 66 cross-linked LEO satellites at 780 km in six near-polar planes cover

the entire Earth, including the poles. They serve handheld phones, IoT devices and aircraft tracking. The second-generation **Iridium NEXT** constellation was completed in 2019. **Globalstar** uses bent-pipe LEO satellites and, since 2022, provides emergency messaging for Apple iPhones.

17.8.2 Direct-to-Device

The newest frontier is connecting **unmodified smartphones** directly to satellites, using the phones' own cellular frequencies. The challenge is the link budget (Figure 17.9): a phone has an antenna gain of about 0 dBi and transmits only about 0.2 W, compared with a broadband terminal's 30–40 dBi. The missing gain must be provided by the satellite — with very large antennas — and by modest data rates.

- **SpaceX** launched its first Direct to Cell satellites in January 2024, and by 2025 several hundred were in orbit. T-Mobile's **T-Satellite** text and data service launched commercially in the USA in July 2025. Partners in Canada, Japan, Australia, Switzerland, Latin America and elsewhere followed. In 2025 SpaceX agreed to acquire mobile-satellite spectrum from EchoStar to expand the service.
- **AST SpaceMobile** builds satellites with enormous phased-array antennas: about 64 m^2 on its first five BlueBirds (2024) and about 223 m^2 on the second-generation BlueBirds. These are the largest commercial communication arrays ever flown. The satellites are designed to deliver broadband speeds of up to about 120 Mbit/s per cell to ordinary phones, in partnership with AT&T, Verizon, Vodafone and others. Thirteen BlueBirds had been launched by August 2026.
- **Lynk Global**, **Skylo** (using existing GEO satellites for messaging) and Chinese providers (Tiantong) offer other approaches. The 3GPP standards now include satellite access, so future phones and networks will treat satellites as just another cell.

17.9 Gateways, Teleports and Networks

Satellites are only part of a communication system. **Gateways** (also called teleports or ground stations) connect the satellite network to the terrestrial internet and telephone networks. A GEO HTS may need 10–50 gateways spread over its coverage area, each with 7–9 m Ka-band antennas and fibre connections, located far enough apart that rain fades are uncorrelated. A LEO constellation needs many more — Starlink operates well over a hundred gateway sites worldwide, each with several tracking antennas — although laser links reduce the need for gateways in remote areas. Network operations centres control the satellites, allocate capacity, manage subscribers and monitor quality of service, and increasingly the whole system is managed as software, with virtualised network functions running in commercial clouds.

Satellite networks are also integrating with terrestrial networks. Mobile operators use satellite **backhaul** to connect cell towers in remote areas. Airlines and shipping companies combine LEO, GEO and cellular links on the same terminal. The 5G non-terrestrial-network standards also allow a satellite to act as part of a mobile operator's radio-access network. The long-term vision is a seamless network in which users neither know nor care whether their data travel through fibre, a cell tower or a satellite.

TABLE 17.3 Selected high-throughput satellites (approximate design capacities).

Satellite (operator)	Launch	Orbit	Throughput	Notes

Satellite (operator)	Launch	Orbit	Throughput	Notes
KA-SAT (Eutelsat)	2010	GEO 9° E	≈ 90 Gbit/s	82 Ka-band spot beams over Europe
ViaSat-1 (Viasat)	2011	GEO	≈ 140 Gbit/s	First HTS above 100 Gbit/s
Intelsat EpicNG series	2016–	GEO	25–60 Gbit/s each	Ku/Ka multi-spot, open architecture
ViaSat-2 (Viasat)	2017	GEO	≈ 260 Gbit/s	Antenna deployment problem reduced capacity
Jupiter 3 / EchoStar XXIV (Hughes)	2023	GEO 95° W	≈ 500 Gbit/s	One of the heaviest commercial GEO satellites (≈ 9 t)
ViaSat-3 F1, F2 (Viasat)	2023, 2025	GEO	designed > 1 Tbit/s each	F1 reflector partially deployed
O3b mPOWER (SES)	2022–	MEO	multi-Tbit/s system	Up to ≈ 5 000 beams per satellite

17.10 Optical Communications and Data Relay

17.10.1 Laser Links

Optical communications offer far higher data rates than radio with much smaller antennas and no spectrum licensing (Chapter 7). Inter-satellite laser links are now routine in LEO constellations: Starlink operates thousands of laser terminals carrying up to about 100 Gbit/s per link. The **European Data Relay System (EDRS)** has relayed Sentinel data from LEO to Europe via GEO at 1.8 Gbit/s since 2016. NASA's **Laser Communications Relay Demonstration (LCRD)** (2021) and the **ILLUMA-T** terminal on the ISS (2023) demonstrated an end-to-end laser relay from the space station through GEO to the ground. The TBIRD CubeSat (2022) downlinked at 200 Gbit/s. The main limitation of space-to-ground laser links is cloud cover, which is overcome with networks of optical ground stations in dry locations.

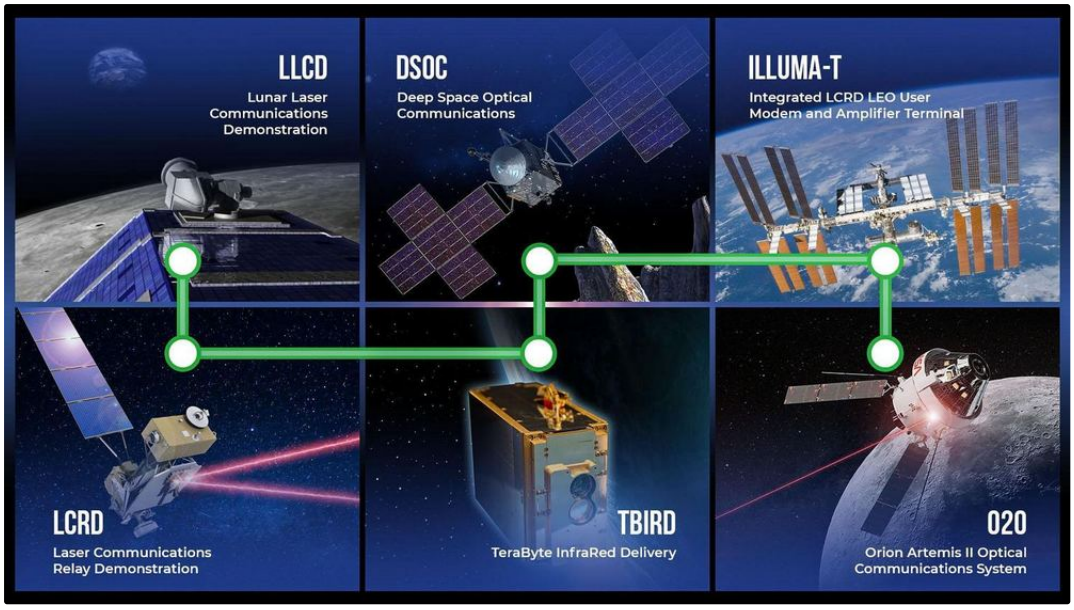


FIGURE 17.17 NASA's optical communications demonstrations: the Lunar Laser Communications Demonstration (2013), Deep Space Optical Communications on Psyche (2023), ILLUMA-T on the ISS, LCRD in geostationary orbit, TBIRD and the Orion Artemis II optical system. Image: NASA/JSC.

17.10.2 Data Relay Satellites

Low-orbiting spacecraft see a given ground station for only about 10 minutes per pass. **Data relay satellites** in GEO provide near-continuous contact. NASA's **Tracking and Data Relay Satellite System (TDRSS)** has operated since 1983. It has supported the Space Shuttle, the ISS, the Hubble Space Telescope and many science missions. TDRS-M, launched in 2017 (the chapter's opening image), was the last of the series. NASA plans to retire TDRS in the 2030s and buy relay services from commercial providers instead, and in 2022 it awarded development agreements to several companies for this purpose. China operates the Tianlian relay satellites and Russia the Luch series; ESA's EDRS uses laser links.

EXAMPLE: HOW MANY SATELLITES DOES A LEO BROADBAND SYSTEM NEED?

Coverage. A satellite at 550 km serving users at elevations above 25° covers a circle of Earth-central half-angle

$$\lambda = \arccos\left(\frac{R_E \cos\varepsilon}{R_E + h}\right) - \varepsilon = \arccos\left(\frac{6371\cos 25^\circ}{6921}\right) - 25^\circ \approx 8.4^\circ$$

so each footprint covers about 2.8 million km² — about 0.54 % of the Earth's surface. Continuous single coverage of the whole Earth therefore needs at least about 190 satellites, and in practice, with overlap and uneven distribution, 300–600.

Capacity. Suppose 10 million subscribers in the busiest regions each need an average of 3 Mbit/s at the evening peak, after statistical multiplexing: a total demand of 30 Tbit/s. Only satellites above populated areas carry this traffic, perhaps 10–15 % of the fleet at any moment. With 100 Gbit/s per satellite, about 300 satellites must be over the busy regions — implying a fleet of 2 000–3 000 satellites. Doubling the subscriber base or the demand per user doubles the fleet unless capacity per satellite grows.

This is why broadband constellations are sized by **capacity**, not coverage, and why capacity per satellite (and the launch vehicle needed to carry larger satellites) is the key to their economics.

AI IN SATELLITE COMMUNICATIONS

Modern satellite networks are too complex to manage by hand. Machine learning predicts traffic demand and rain fades, allocates beams, frequencies and power dynamically on software-defined satellites, routes traffic through laser meshes, detects interference and jamming, and plans collision-avoidance manoeuvres for thousands of satellites. Chapter 49 discusses AI for satellite operations and networks.

KEY TAKEAWAYS

- Communication satellites evolved from passive balloons (Echo, 1960) through GEO trunking and broadcasting to high-throughput GEO satellites and LEO broadband constellations.
- Orbit choice trades coverage and simplicity (GEO) against latency and terminal size (LEO); MEO and HEO fill specific niches.
- Higher frequencies give more bandwidth but suffer rain fade; systems use adaptive coding and modulation, power control and site diversity.
- The ITU allocates spectrum and coordinates orbital positions; EPFD limits protect GEO from LEO interference; milestone rules require constellations to be deployed on time.
- Capacity comes from frequency reuse with many spot beams; HTS increased GEO capacity per satellite to 0.5–1 Tbit/s.
- Starlink (about 11 000 satellites in October 2026 and about 12 million customers) has transformed the market; OneWeb, Amazon Leo, Guowang, Qianfan and others follow.
- Direct-to-device services connect ordinary phones by using very large antennas in orbit and 3GPP non-terrestrial-network standards.
- Laser links are now routine between satellites; data-relay satellites keep low-orbit spacecraft in near-continuous contact.

REVIEW QUESTIONS

1. Trace the evolution of communication satellites from Echo to LEO broadband constellations and give one key innovation at each stage.
2. Calculate the one-way propagation delay of a link through a geostationary satellite (two legs of 38 000 km) and through a 550 km LEO satellite (two legs of 1 000 km).
3. Why does rain fade increase with frequency? Describe three techniques used to keep Ka-band links available.
4. What are EPFD limits and ITU milestone rules, and why do they matter for new constellations?
5. Compare bent-pipe and regenerative transponders.
6. Explain how frequency reuse with many spot beams increases the capacity of a high-throughput satellite.
7. Why do direct-to-device services need very large antennas in orbit?
8. How do laser inter-satellite links and data-relay satellites change the design of a LEO constellation's ground segment?

CHAPTER

18 Global Navigation Satellite Systems

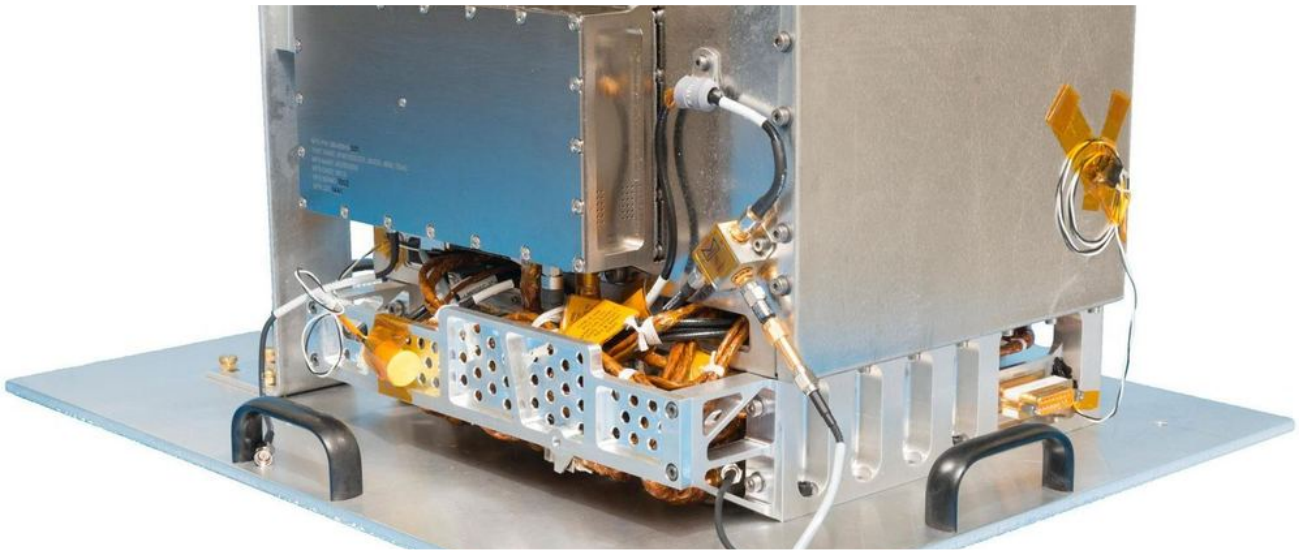


Image: NASA/JPL (NASA/JPL-Caltech)

Global navigation satellite systems (GNSS) — GPS, GLONASS, Galileo and BeiDou, with regional systems in Japan and India — broadcast precisely timed radio signals from atomic clocks in orbit, allowing billions of receivers to determine their position, velocity and time anywhere on Earth. This chapter explains how satellite navigation works: one-way ranging, pseudoranges and the navigation solution; spread-spectrum signals and receivers; atomic clocks and relativity; the error budget and satellite geometry; and the augmentation techniques that improve accuracy from metres to centimetres. It describes the four global systems as they stand in 2026, the enormous range of applications that depend on them — including the timing of telecom networks, power grids and financial markets — and their vulnerability to interference, jamming and spoofing, which has grown into a serious problem. It ends with the next generation of satellites and with navigation in low Earth orbit and on the Moon.

IN THIS CHAPTER

18.1 Positioning, Navigation and Timing
18.2 How Satellite Navigation Works
18.3 The Systems in 2026
18.4 Accuracy
18.5 Augmentation: From Metres to Centimetres
18.6 Applications
18.7 Vulnerabilities: Interference, Jamming and Spoofing
18.8 Future Systems

18.1 Positioning, Navigation and Timing

Satellite navigation is one of the most pervasive technologies ever created. In 2026 there are roughly seven billion GNSS receivers in use — in phones, vehicles, aircraft, ships, tractors, drones, watches, survey instruments and the timing equipment of telecommunications networks, power grids and stock exchanges. Most users think of GNSS as a way to find their position, but its three outputs — **positioning, navigation and timing (PNT)** — are equally important. Precise time from GNSS synchronises mobile-phone base stations, time-stamps financial transactions and aligns measurements across electricity grids. A study for the US government estimated that GPS alone generated about $1.4 trillion in economic benefits in the USA. This figure covers the years between its opening to civil use in the 1980s and 2017.

18.1.1 A Short History

The idea grew from the space race. When Sputnik was launched in 1957, scientists at the Johns Hopkins Applied Physics Laboratory found that they could determine its orbit from the Doppler shift of its radio signal. They then realised that, conversely, a receiver could determine its own position from the Doppler shift of a satellite in a known orbit. The resulting US Navy **Transit** system (first satellite 1960, operational 1964) allowed submarines to fix their position every hour or so. Experiments with atomic clocks in orbit (Timation) and the Air Force's 621B programme led in 1973 to the decision to build the **NAVSTAR Global Positioning System**. The first GPS satellite was launched in 1978 and the system was declared fully operational in 1995. The Soviet Union developed the similar **GLONASS** from 1982.

GPS was opened to civil users in the 1980s, but its civil signal was deliberately degraded by "Selective Availability" to about 100 m accuracy. This degradation was switched off in May 2000, instantly improving accuracy tenfold and unleashing the consumer navigation industry. Other powers then built their own systems for strategic independence: China's **BeiDou** (regional from 2012, global in 2020), Europe's civilian **Galileo** (initial services in 2016), Japan's regional **QZSS** and India's regional **NavIC** (Figure 18.1).

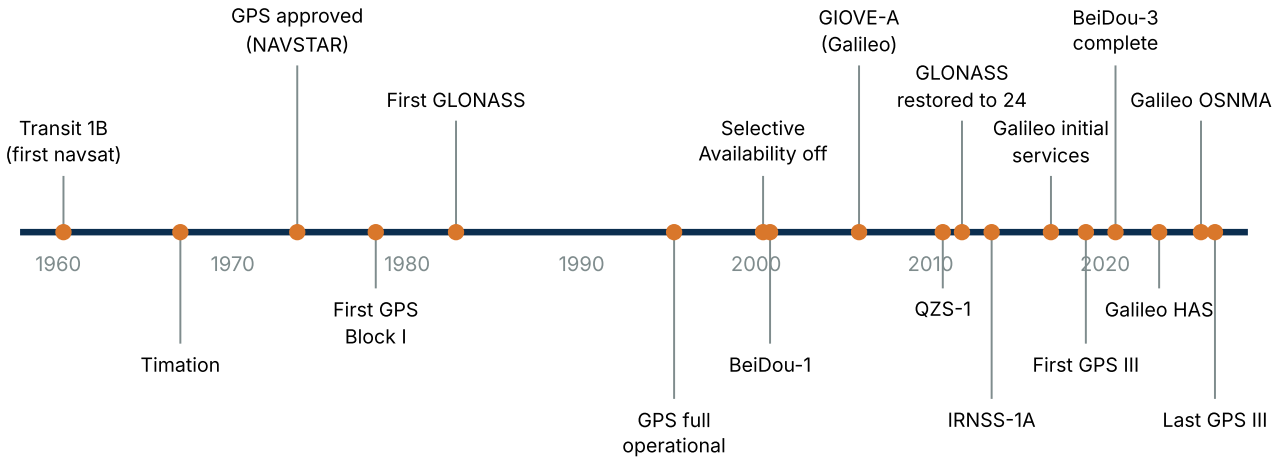


FIGURE 18.1 Milestones in satellite navigation.

18.2 How Satellite Navigation Works

18.2.1 One-Way Ranging and Pseudoranges

Each GNSS satellite carries atomic clocks and continuously broadcasts radio signals that contain its precise time and its position (ephemeris). A receiver measures when each signal arrives; the difference between the time of reception and the time of transmission, multiplied by the speed of light, is the distance (range) to the satellite. With ranges to three satellites at known positions, the receiver's position could be found by **trilateration** — the intersection of three spheres (Figure 18.2).

The difficulty is that the receiver's own clock — a cheap quartz oscillator — is not synchronised with the satellites' atomic clocks. An error of just one microsecond would cause a range error of 300 m. The measured ranges, which include this unknown clock offset, are therefore called **pseudoranges**:

$$\rho_i = \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2} + c\,\delta t + \varepsilon_i$$

where (x_i, y_i, z_i) is the position of satellite i, (x, y, z) the receiver position, δt the receiver clock offset and ε_i the remaining errors. Because the clock offset is common to all measurements, it can be solved as a fourth unknown: **four satellites** are needed to determine three position coordinates and time. In practice receivers track 10–40 satellites from several constellations and solve the over-determined system by iterative least squares or, for continuous navigation, by a Kalman filter. As a by-product, every GNSS receiver becomes a clock synchronised to the system time within tens of nanoseconds. Velocity is found in a similar way from the Doppler shifts of the signals.

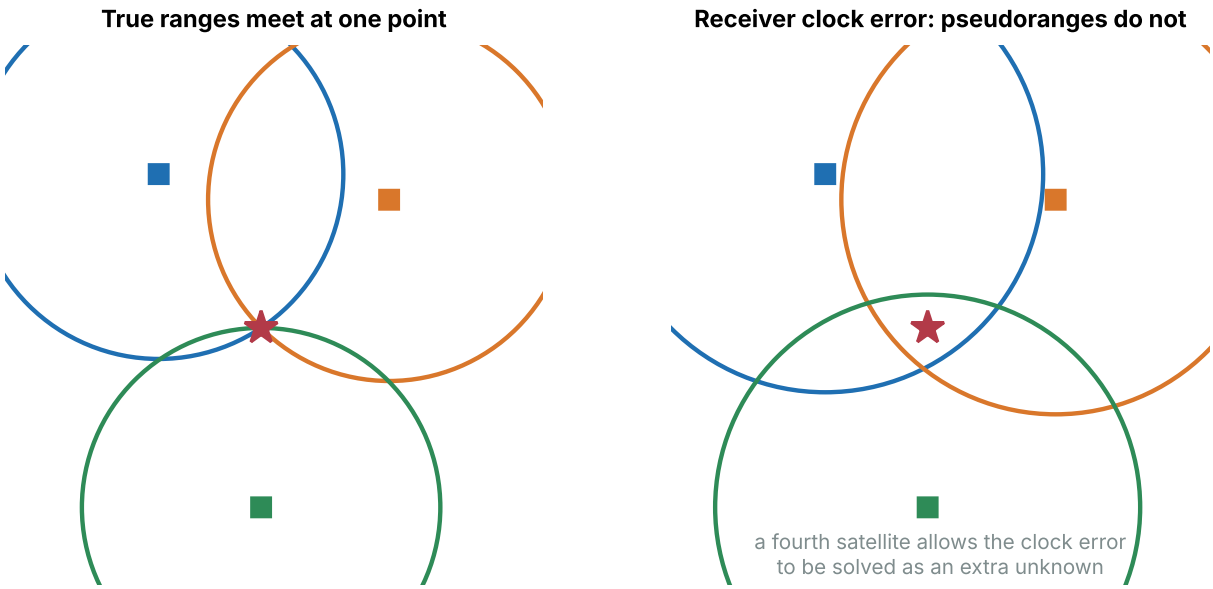
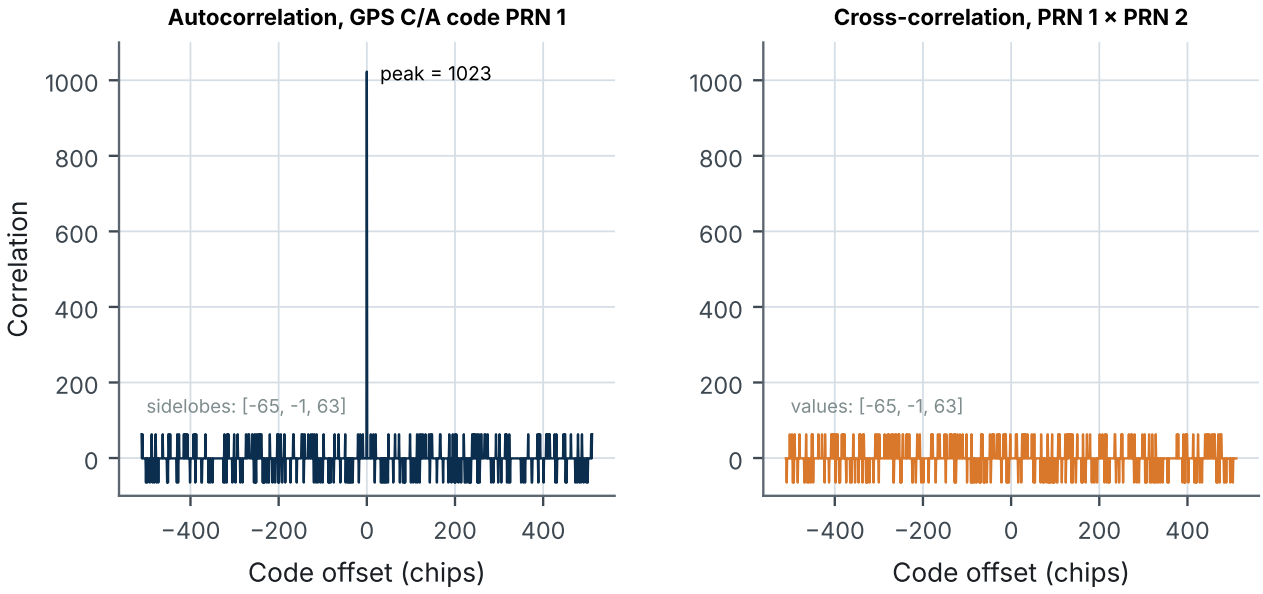


FIGURE 18.2 Trilateration. With perfect clocks, ranges from three satellites intersect at the user's position (left). A receiver clock error changes all ranges by the same amount, so the circles no longer meet (right) — the error is solved for using a fourth satellite.

18.2.2 Spread-Spectrum Signals

GNSS signals are extraordinarily weak. A GPS L1 signal arrives at the Earth's surface with a minimum power of about −128.5 dBm (1.4×10^{-16} W). That is roughly the brightness of a 25 W light bulb seen from 20 000 km, and well **below** the thermal noise in the receiver's bandwidth. Receivers can nevertheless detect them because the signals use **direct-sequence spread spectrum**: the navigation data are multiplied by a fast **pseudo-random noise (PRN) code** that spreads the signal over a wide bandwidth. The receiver generates a replica of each satellite's code and correlates it with the incoming signal; only when the replica is aligned with the received code does a strong correlation peak appear (Figure 18.3). Correlating over 1–20 ms or more provides a **processing gain** of 30–45 dB that lifts the signal out of the noise.

Each satellite uses a different code chosen to have low **cross-correlation** with the others, so all satellites can transmit on the same frequency without interfering — **code division multiple access (CDMA)**. The GPS **coarse/acquisition (C/A) code**, for example, is a Gold code of 1 023 chips transmitted at 1.023 million chips per second, repeating every millisecond. The timing of the correlation peak, refined by tracking loops to a small fraction of a chip, gives the pseudorange. The phase of the carrier wave (wavelength 19 cm at L1) can be measured even more precisely, to about a millimetre.



Computed with the IS-GPS-200 G1/G2 Gold-code generators (1 023 chips, 1.023 Mchip/s).

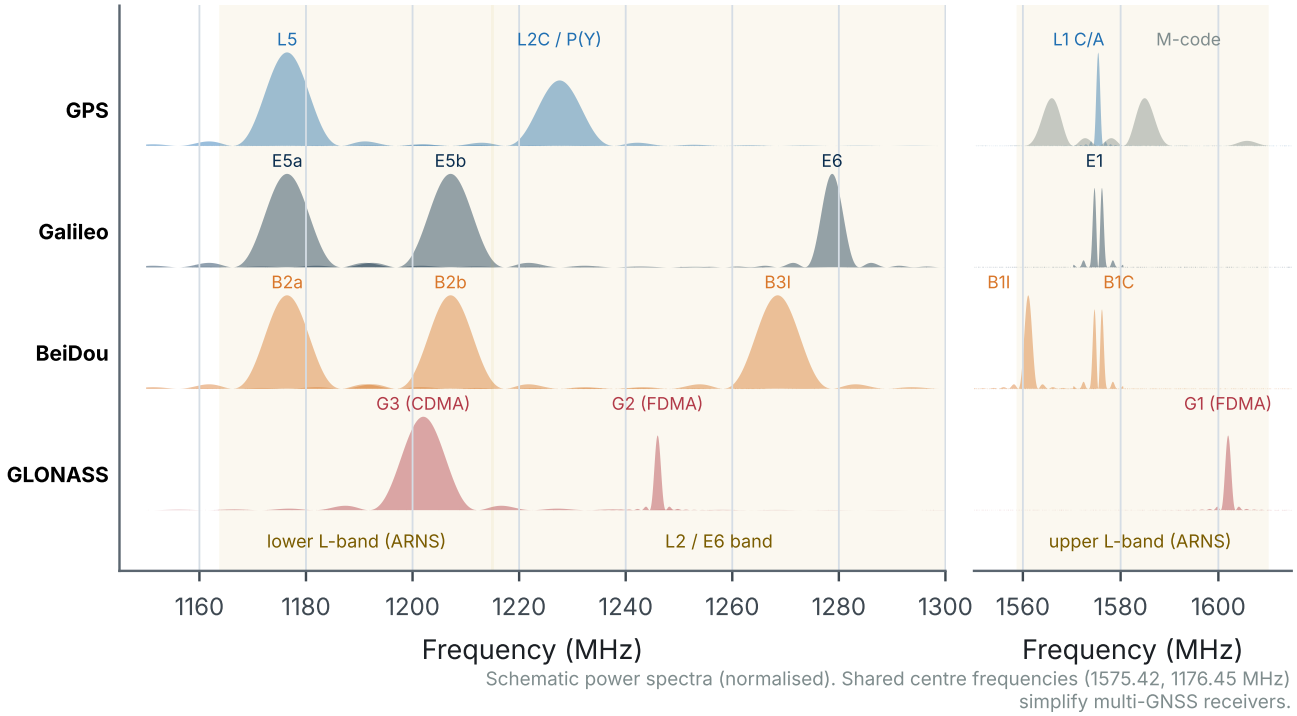
FIGURE 18.3 Correlation properties of the GPS C/A code, computed from the generators in the GPS interface specification. The autocorrelation has a single sharp peak of 1 023 at zero offset; sidelobes and cross-correlations with other satellites' codes take only the values −65, −1 and +63.

18.2.3 Receivers

A GNSS receiver consists of an antenna (typically right-hand circularly polarised), a radio-frequency front end that amplifies, filters and down-converts the signals and digitises them, and a baseband processor with many **correlator channels**. **Acquisition** searches over all possible code delays and Doppler shifts for each satellite. Modern receivers perform this two-dimensional search in parallel using FFTs. It takes from about a second (with assistance data from the mobile network) to a minute (cold start). **Tracking** then follows each signal with a delay-locked loop for the code and a phase- or frequency-locked loop for the carrier, while the navigation message is decoded. A smartphone GNSS chip, which tracks four constellations on two frequencies, costs about a dollar and consumes tens of milliwatts.

18.2.4 Signals of the Four Global Systems

All four global systems now broadcast multiple civil signals in the radio-navigation bands around 1.2 and 1.575 GHz, which are protected for aeronautical safety use (Figure 18.4). Newer signals share common centre frequencies: 1575.42 MHz (GPS L1C, Galileo E1, BeiDou B1C) and 1176.45 MHz (GPS L5, Galileo E5a, BeiDou B2a). This allows a single dual-frequency receiver to use all of them. Some use **binary offset carrier (BOC)** modulation, which splits the signal's power away from the band centre, improving tracking precision and reducing interference with older signals.



Schematic power spectra (normalised). Shared centre frequencies (1575.42, 1176.45 MHz) simplify multi-GNSS receivers.

FIGURE 18.4 Simplified frequency plan of the four global navigation systems.

TABLE 18.1 Principal signals of the global navigation satellite systems (simplified).

System	Signal	Centre frequency (MHz)	Modulation	Use
GPS	L1 C/A	1575.42	BPSK(1)	Legacy civil signal, used by almost every receiver
GPS	L1C	1575.42	TMBOC(6,1,1/11)	Modern civil, interoperable with Galileo E1 and BeiDou B1C (GPS III)
GPS	L2C	1227.60	BPSK(1) time-multiplexed	Second civil frequency (since 2005)
GPS	L5	1176.45	BPSK(10)	Safety-of-life civil signal (since 2010)
GPS	P(Y), M-code	1575.42, 1227.60	BPSK(10), BOC(10,5)	Military, encrypted
Galileo	E1 OS	1575.42	CBOC(6,1,1/11)	Open service; carries OSNMA authentication
Galileo	E5a, E5b	1176.45, 1207.14	AltBOC(15,10) combined	Open service; E5b also for integrity
Galileo	E6	1278.75	BPSK(5)	High-Accuracy Service; Public Regulated Service (encrypted)
BeiDou	B1I, B1C	1561.098, 1575.42	BPSK(2), MBOC	Open service
BeiDou	B2a, B2b	1176.45, 1207.14	QPSK(10)	Open service; B2b carries PPP corrections
BeiDou	B3I	1268.52	BPSK(10)	Open and authorised services

System	Signal	Centre frequency (MHz)	Modulation	Use
GLONASS	L1OF, L2OF	1602 + k·0.5625, 1246 + k·0.4375	FDMA, BPSK(0.511)	Legacy open signals, one frequency channel per satellite pair
GLONASS	L3OC	1202.025	CDMA	New CDMA signal (GLONASS-K)

18.3 The Systems in 2026

18.3.1 Space Segments

TABLE 18.2 The world's satellite navigation systems in 2026.

System	Operator	Nominal constellation	MEO orbit	Other orbits	Clocks	Status (Oct 2026)
GPS (USA)	US Space Force	24 slots (≈ 31 operational)	20 180 km, 55°, 6 planes	—	Rubidium, caesium	Operational since 1995; ten GPS III launched 2018–2026; GPS IIIF from ≈ 2027
GLONASS (Russia)	Roscosmos / Ministry of Defence	24	19 100 km, 64.8°, 3 planes	—	Caesium, rubidium	Operational; transition to GLONASS-K/K2 with CDMA signals
Galileo (EU)	EU / EUSPA, ESA	24 + spares	23 222 km, 56°, 3 planes	—	Passive hydrogen maser, rubidium	Initial services since 2016; ≈ 30 satellites usable after the December 2025 launch; second generation from 2026–27
BeiDou-3 (China)	China Satellite Navigation Office	24 MEO + 3 IGSO + 3 GEO	21 528 km, 55°, 3 planes	GEO and inclined GEO over Asia	Rubidium, hydrogen maser	Global service since 2020; legacy BeiDou-2 satellites still in use
QZSS (Japan)	Cabinet Office	7 (growing to 11)	—	Quasi-zenith (inclined, elliptical GEO) and GEO	Rubidium	Six satellites by October 2026
NavIC (India)	ISRO	7	—	GEO and IGSO over India	Rubidium	Several clock failures; NVS-series replacements, one (NVS-02) stranded in transfer orbit in 2025

The global systems use medium Earth orbits at 19 000–23 000 km (Figure 18.5). This is a compromise between coverage (each satellite sees about 38 % of the Earth) and the number of satellites needed. About 24 satellites give at least four visible everywhere at all times. GLONASS uses a higher inclination (64.8°) for better coverage of Russia's northern latitudes. BeiDou and the regional systems add satellites in geostationary and **inclined geosynchronous (IGSO)** orbits. These trace figure-of-eight ground tracks and remain over their service regions. Japan's quasi-zenith satellites are designed to stay at high elevation over Japan, so that their signals are not blocked by tall buildings and mountains.

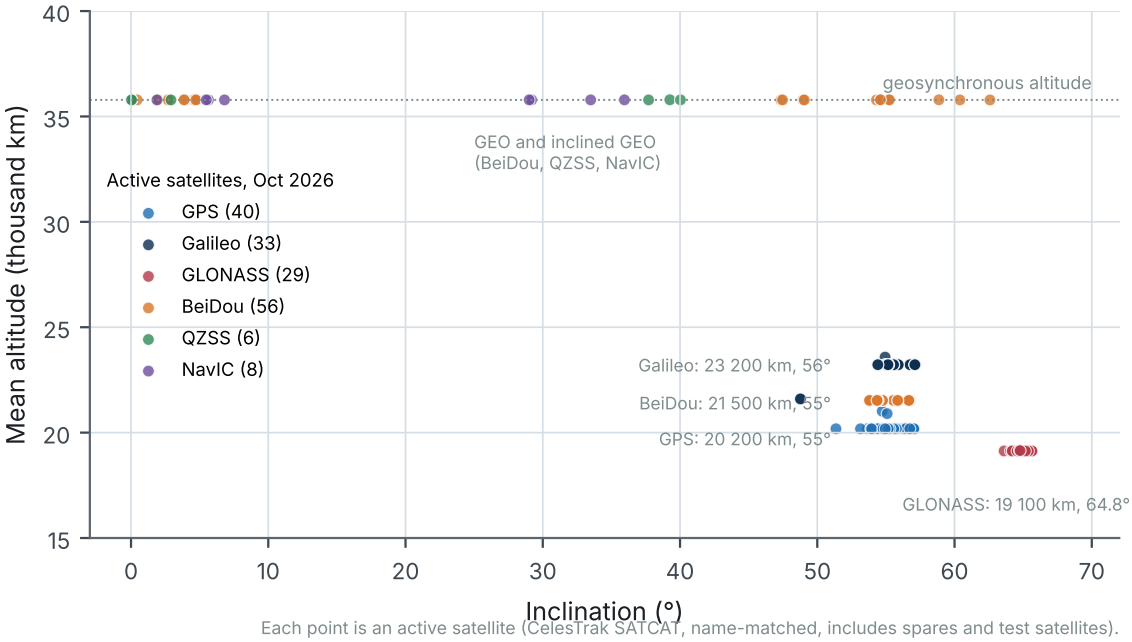


FIGURE 18.5 Altitude and inclination of active navigation satellites in October 2026. Each global system occupies a distinct MEO shell; BeiDou, QZSS and NavIC also use geostationary and inclined geosynchronous orbits.



FIGURE 18.6 A Delta II rocket ready to launch the 19th NAVSTAR GPS satellite from Cape Canaveral in 1993. Delta II rockets launched most GPS satellites from 1989 to 2009. Image: NASA/MSFC.

18.3.2 Atomic Clocks and Time

The precision of GNSS rests on atomic clocks. GPS satellites carry rubidium and caesium frequency standards. Galileo satellites carry two rubidium clocks and two **passive hydrogen masers**. These masers are the most stable clocks flying in navigation satellites, with a stability of better than one part in 10^{14} over a day. BeiDou-3 also carries hydrogen masers. A clock error of 1 ns produces a range error of 30 cm, so the clocks must be extremely stable between updates from the ground. Each system maintains its own **system time** (GPS Time, Galileo System Time, BeiDou Time, GLONASS Time), steered to Coordinated Universal Time (UTC). The offsets between them are broadcast, so that receivers can combine all systems. NASA's **Deep Space Atomic Clock** (the opening image of this chapter), a mercury-ion clock flown in 2019–2021, demonstrated stability good enough for spacecraft to navigate autonomously in deep space.

18.3.3 Relativity in Daily Life

GNSS is the most familiar practical application of Einstein's theories of relativity (Figure 18.7). A clock in a GPS orbit runs **faster** than a clock on the ground by about 45.7 µs per day. This is because it sits higher in the Earth's gravitational field (general relativity). It also runs **slower** by about 7.2 µs per day because of its orbital speed (special relativity). The net effect is +38.5 µs per day — which, left uncorrected, would cause range errors growing by about 11 km per day. GPS satellite clocks are therefore deliberately set to run slow by 4.465 parts in 10^{10} before launch, and receivers apply a further small correction for orbital eccentricity.

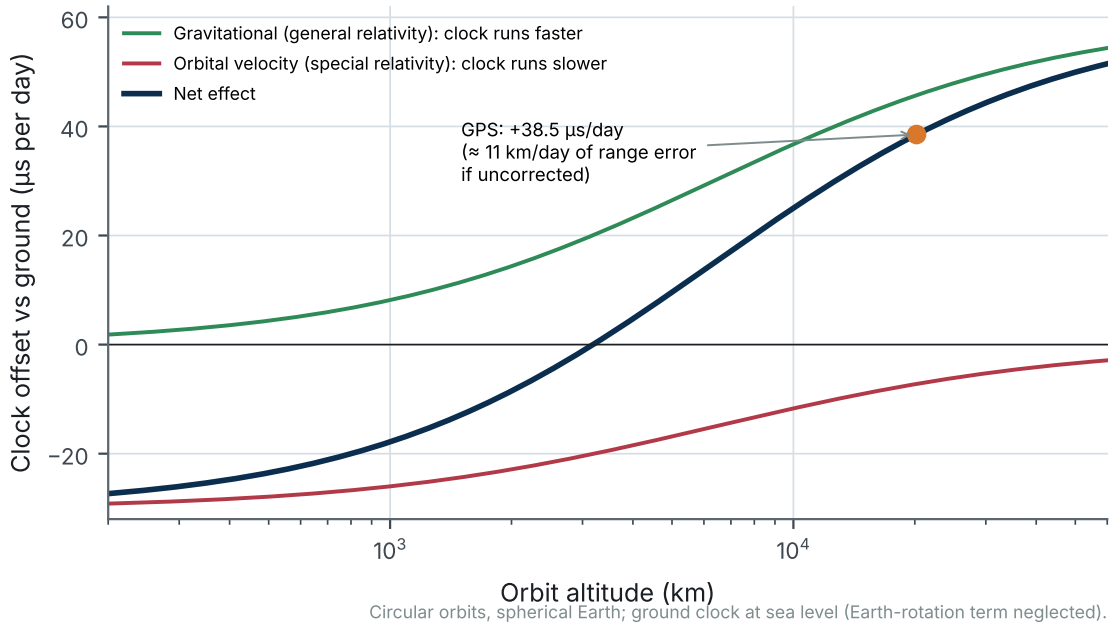


FIGURE 18.7 Relativistic clock-rate offsets of orbiting clocks relative to a clock on the ground. Above about 3 000 km the gravitational effect dominates and orbiting clocks gain time.

18.3.4 Control Segments

Each system has a control segment of monitor stations around the world that track the satellites, and master control stations that compute orbit and clock predictions. The segment's uplink antennas upload these predictions to the satellites for broadcast. The GPS master control station at Schriever Space Force Base in Colorado updates each satellite's navigation message at least

daily. The modernised GPS ground system, **OCX**, is needed to exploit the new civil and military signals fully. It suffered years of delays and cost growth. Galileo's ground segment includes control centres in Germany and Italy and a worldwide network of sensor stations.

18.4 Accuracy

18.4.1 The Error Budget

The accuracy of a GNSS position depends on the errors in each pseudorange — summarised as the **user equivalent range error (UERE)** — and on the geometry of the satellites. The main error sources are (Figure 18.8):

- **Satellite clock and orbit errors**: the broadcast predictions are not perfect. The GPS signal-in-space range error has improved to about 0.5 m (RMS) in recent years, and Galileo's to about 0.2 m.
- **Ionospheric delay**: free electrons in the ionosphere (50–1 000 km) delay the signal. The delay is proportional to the **total electron content (TEC)** along the path and inversely proportional to the square of the frequency (Figure 18.9). It is typically 2–15 m at L1, and much more during solar maxima and geomagnetic storms (Chapter 22). Single-frequency receivers remove only about half of it using a broadcast model; **dual-frequency** receivers eliminate the first-order effect almost completely, which is the main reason for the new civil signals.
- **Tropospheric delay**: the neutral atmosphere delays signals by about 2.4 m at the zenith (more at low elevations); models remove most of it, leaving errors of decimetres.
- **Multipath**: reflections from buildings, the ground and vehicles distort the correlation peak — the dominant error in cities.
- **Receiver noise**, typically decimetres for code measurements and millimetres for carrier phase.

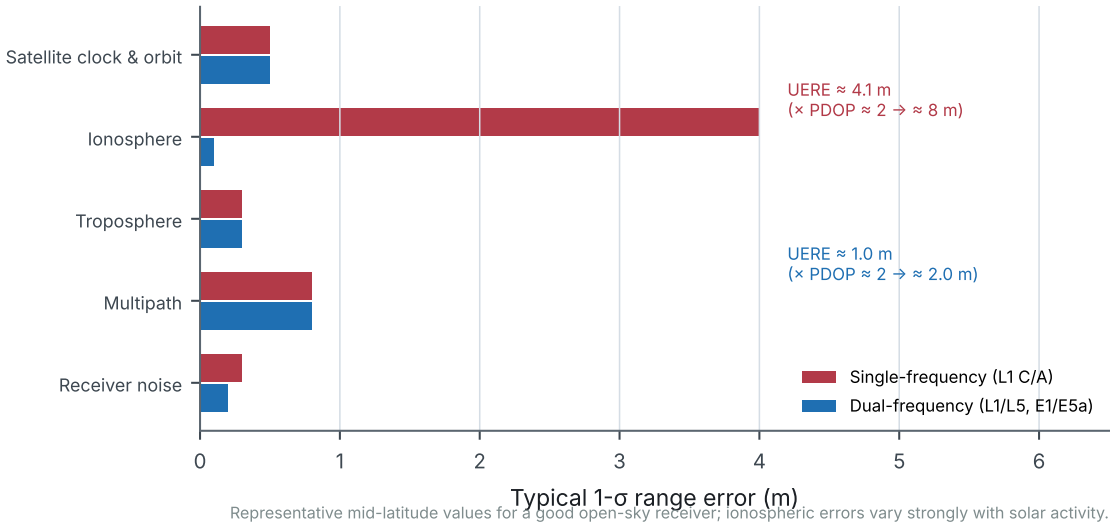


FIGURE 18.8 Typical pseudorange error budget for single- and dual-frequency receivers. Dual-frequency operation removes the ionospheric error, the largest error source for single-frequency users.

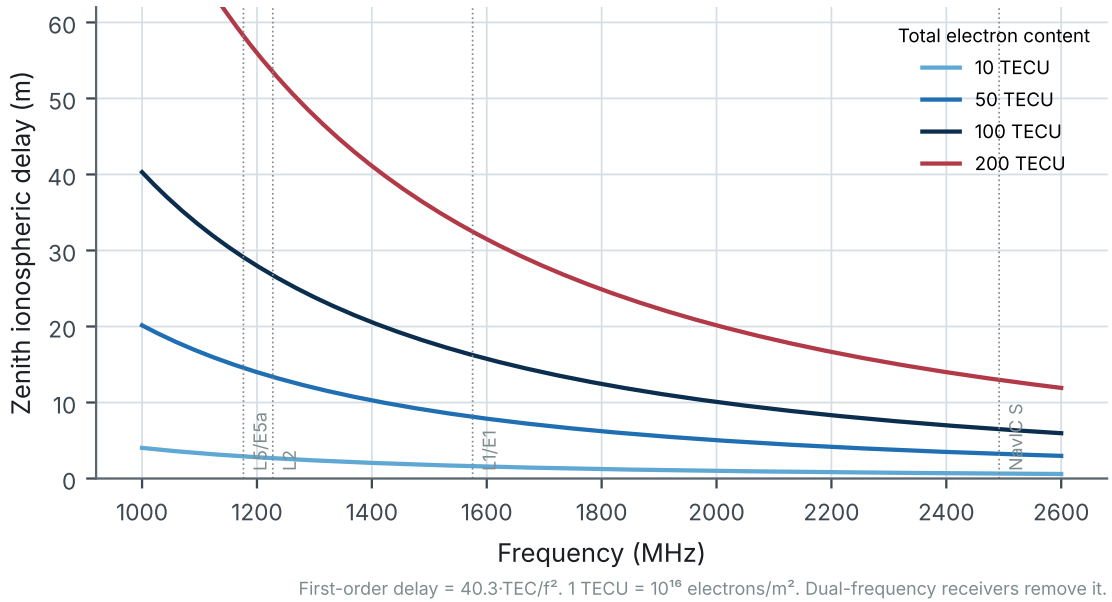


FIGURE 18.9 Ionospheric delay versus frequency for different total electron contents. The delay at L1 can exceed 30 m in a strongly ionised ionosphere.

18.4.2 Satellite Geometry and DOP

The same range errors produce larger position errors when the satellites are poorly spread across the sky (Figure 18.10). This amplification is expressed by the **dilution of precision (DOP)**, computed from the geometry matrix of the least-squares solution. Position error ≈ PDOP × UERE (with HDOP and VDOP for horizontal and vertical components, and TDOP for time). Vertical errors are typically 1.5–2 times larger than horizontal errors because all satellites are above the horizon.

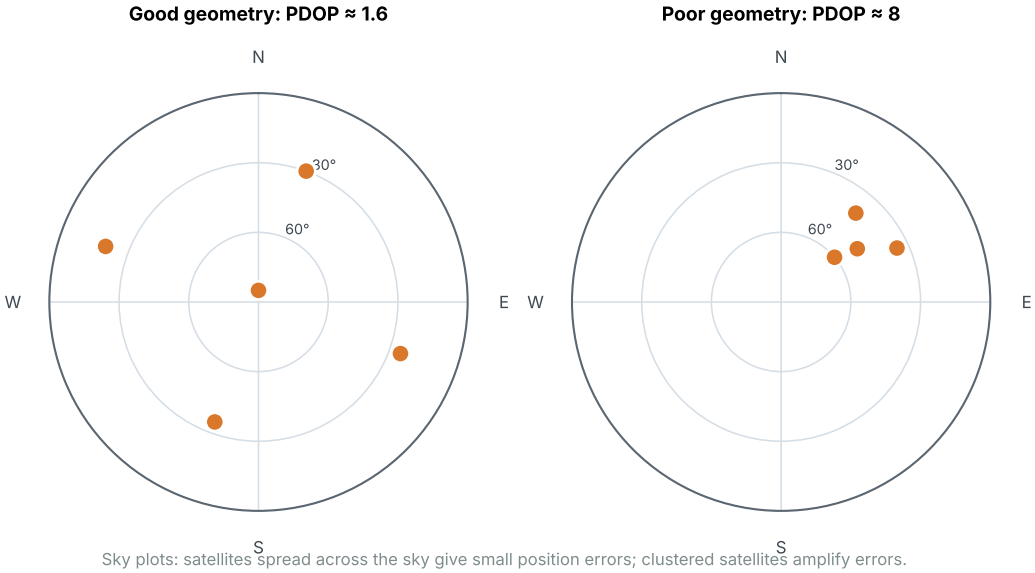
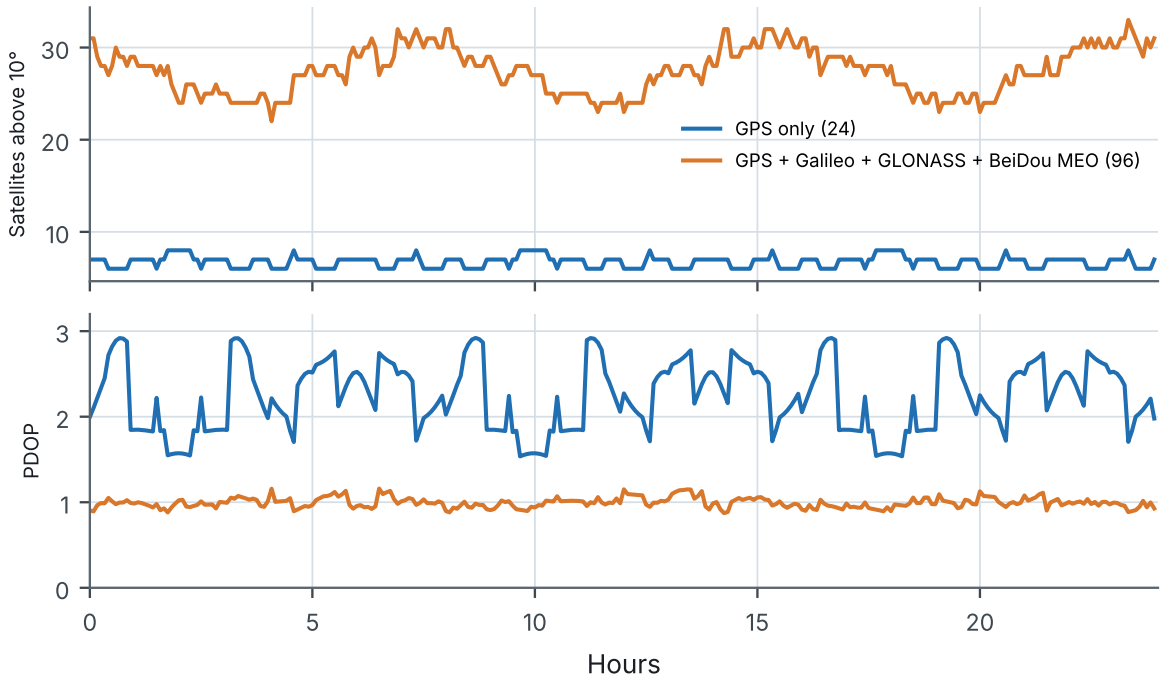


FIGURE 18.10 Satellite sky plots illustrating good and poor geometry. Clustered satellites give a large dilution of precision.

Using several constellations together dramatically improves geometry and availability (Figure 18.11): a multi-GNSS receiver typically sees 30–40 satellites rather than 8–12, and PDOP falls from about 2 to about 1. This matters most where much of the sky is blocked — in city “urban canyons”, forests and mountains.



Simulated nominal constellations (Walker patterns) seen from 40° N, 0° E, 10° elevation mask.

FIGURE 18.11 Simulated number of visible satellites and PDOP over 24 hours for GPS alone and for four global constellations together.

A typical smartphone achieves 3–5 m accuracy in open sky (dual-frequency phones, introduced in 2018, do better) and worse in cities. A single-frequency GPS receiver meets the official GPS performance standard of a few metres. A dual-frequency, multi-constellation receiver achieves about 1–2 m.

18.5 Augmentation: From Metres to Centimetres

Many applications need more accuracy or guaranteed **integrity** — assurance that the position is not wrong without warning. **Augmentation** systems provide corrections and integrity information (Figure 18.12).

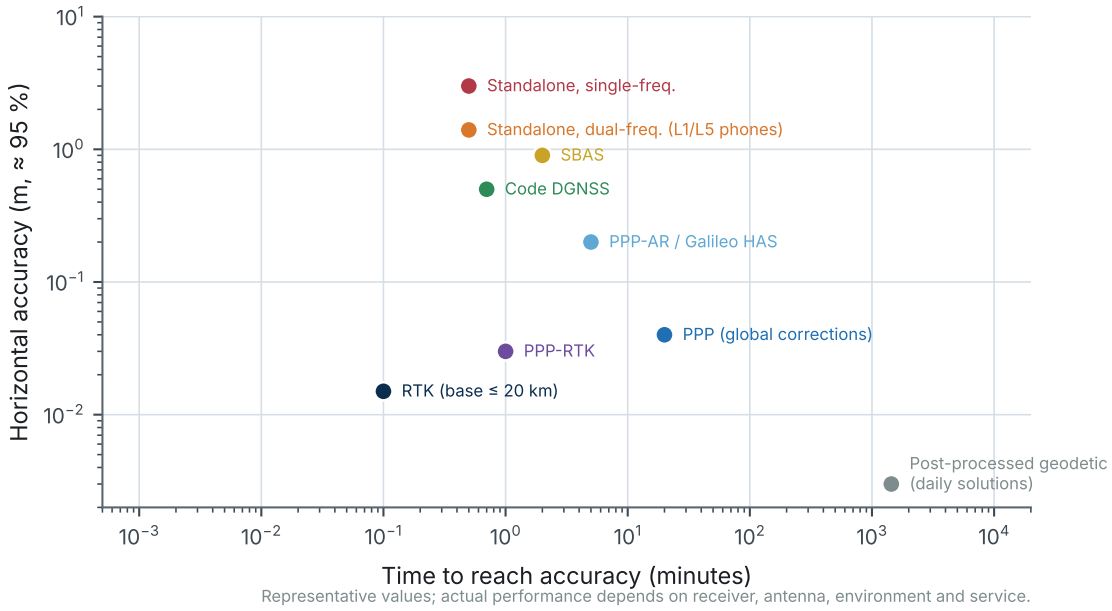
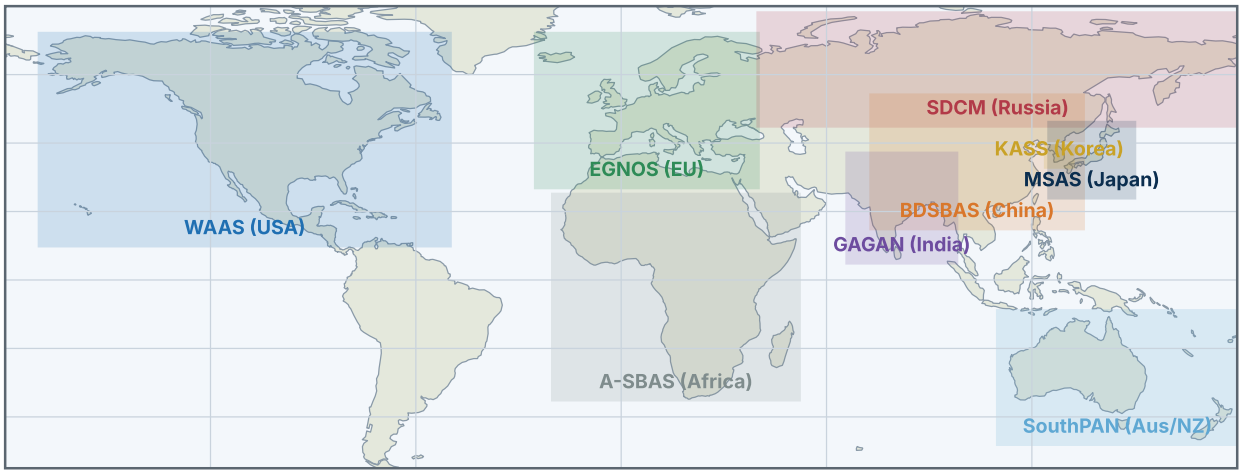


FIGURE 18.12 Typical accuracy versus time to reach it for different GNSS positioning techniques.

- **Differential GNSS (DGNSS)**: a reference station at a known position computes the errors in each pseudorange and broadcasts corrections; nearby users, who experience nearly the same errors, apply them to obtain sub-metre accuracy.
- **Satellite-based augmentation systems (SBAS)** use networks of reference stations across a continent to compute corrections for satellite orbits, clocks and the ionosphere, plus integrity messages. They broadcast them from geostationary satellites on the GPS L1 frequency (Figure 18.13). The US **WAAS** (2003), Europe's **EGNOS** (2009–2011), Japan's MSAS, India's GAGAN and others allow aircraft to fly precision approaches with vertical guidance (LPV) to thousands of airports without ground-based landing systems. The next generation, **dual-frequency multi-constellation (DFMC) SBAS**, will use L1/L5 and Galileo signals.
- **Ground-based augmentation systems (GBAS)** provide corrections and integrity within about 40 km of an airport for Category I (and in future Category II/III) landings.
- **Real-time kinematic (RTK)** positioning uses carrier-phase measurements from a nearby base station (Figure 18.14). Once the integer number of carrier cycles (the **ambiguity**) has been resolved, the rover's position is known to 1–2 cm within seconds. Networks of continuously operating reference stations (CORS) provide RTK corrections across whole countries for surveying, construction and precision agriculture.
- **Precise point positioning (PPP)** uses precise satellite orbit and clock products computed from global networks (such as those of the International GNSS Service). It achieves centimetre accuracy with a single receiver anywhere, after a convergence time of 10–30 minutes. **PPP-RTK** adds regional atmospheric corrections to shorten convergence. Satellite systems now broadcast PPP corrections free of charge: Galileo's **High Accuracy Service (HAS)** since 2023, QZSS's CLAS and MADOCA, and BeiDou's PPP-B2b.



Approximate service regions of satellite-based augmentation systems (SBAS) operating or in development.

FIGURE 18.13 Approximate service regions of satellite-based augmentation systems.

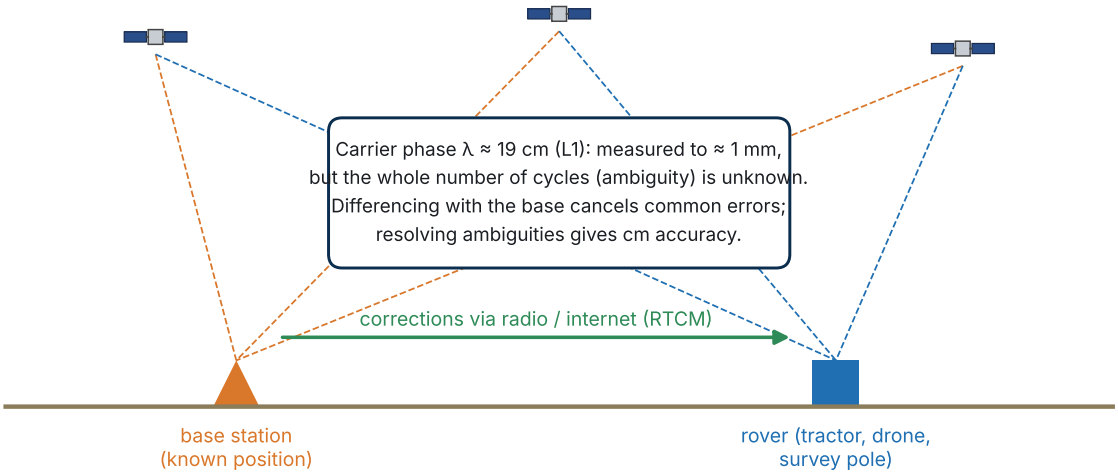


FIGURE 18.14 Real-time kinematic positioning. A base station at a known position sends its carrier-phase measurements to the rover, which can then resolve the carrier-cycle ambiguities and position itself to about a centimetre.

18.6 Applications

TABLE 18.3 Major applications of GNSS.

Sector	Applications	Typical requirement
Consumer and location-based services	Smartphones, maps, ride-hailing, fitness, games	3–5 m, low power, fast fix
Road transport	Navigation, fleet management, emergency call (eCall), tolling, connected and automated vehicles	1–5 m; lane-level 0.5 m for automation, with integrity
Aviation	Performance-based navigation, precision approaches (SBAS, GBAS), ADS-B surveillance	Certified accuracy and integrity
Maritime	Navigation, port approaches, AIS, dynamic positioning of drilling ships	1–10 m; decimetres for dynamic positioning
Rail	Train location, signalling (ERTMS), asset management	Safety-critical integrity
Agriculture	Auto-steering, variable-rate application, yield mapping	2 cm (RTK)

Sector	Applications	Typical requirement
Surveying, construction, mapping	Cadastral surveys, machine control, GIS, mobile mapping	mm–cm
Timing and synchronisation	Telecom networks (5G needs ±1.5 µs), power-grid phasor measurements, financial time-stamping, data centres, broadcasting	10 ns–1 µs
Drones and robotics	Autonomous flight, delivery, inspection	cm–m, high integrity
Search and rescue	Cospas-Sarsat MEOSAR distress beacons; Galileo Return Link Service confirms receipt	< 100 m, minutes
Science	Geodesy and plate tectonics (mm/year), sea level and ice, seismology, ionospheric and atmospheric sounding, reflectometry	mm, ns
Space	Orbit determination of LEO satellites, launch-vehicle tracking, GEO and lunar navigation	cm–m
Defence	Navigation, weapon guidance, timing of military networks	Encrypted, jam-resistant signals

Scientific uses are remarkable. Networks of thousands of geodetic GNSS stations measure the motion of tectonic plates at millimetres per year. They also measure the elastic rebound of the crust as ice sheets melt, and the ground deformation of earthquakes in real time, which can improve tsunami warnings. The same signals measure the atmosphere: water vapour from tropospheric delay, electron content from ionospheric delay (Figure 18.15) and temperature profiles from radio occultation (Chapter 16).

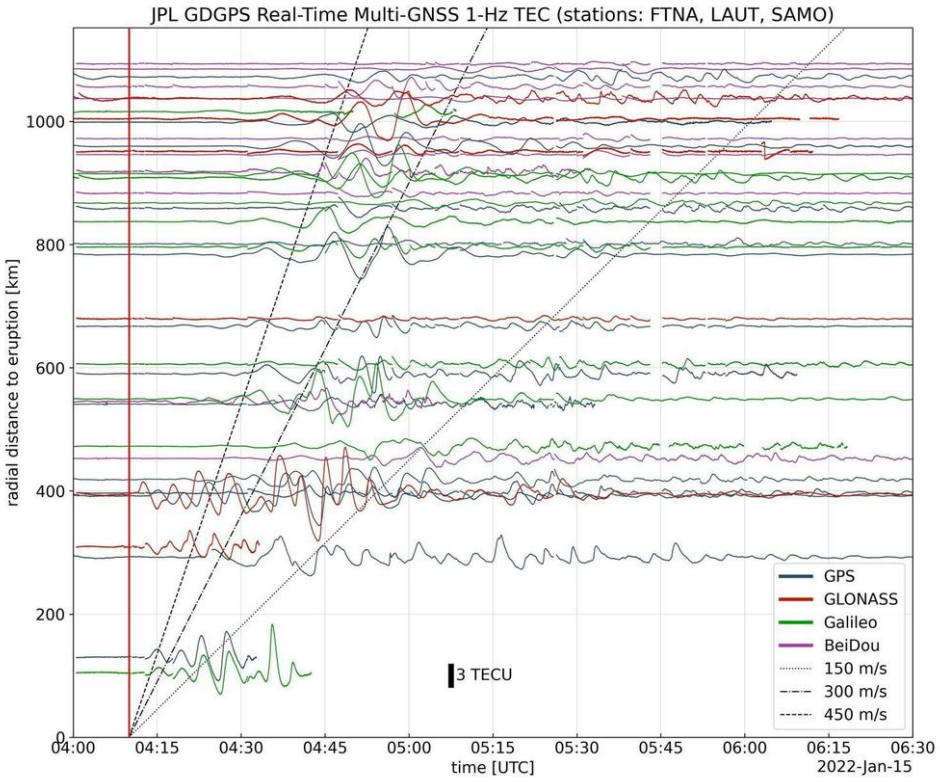


FIGURE 18.15 Ionospheric total electron content measured in real time by NASA's Global Differential GPS network after the eruption of the Hunga Tonga–Hunga Ha'apai volcano on 15 January 2022. Waves travelling outward from the eruption appear as diagonal stripes, measured simultaneously with GPS, GLONASS, Galileo and BeiDou signals. Image: NASA/JPL (NASA/JPL-Caltech/GDGPS).

18.7 Vulnerabilities: Interference, Jamming and Spoofing

The weakness of GNSS signals makes them easy to disrupt. **Jamming** — transmitting noise in the GNSS bands — can deny service over large areas with modest power (Figure 18.16). A 1 W jammer can prevent a typical receiver from tracking GPS C/A signals at line-of-sight distances of more than 10 km. Military jammers of hundreds of watts affect aircraft hundreds of kilometres away. Cheap illegal "privacy jammers" in vehicles regularly disrupt airports and port operations. **Spoofing** — transmitting counterfeit GNSS signals — is more insidious: a spoofed receiver reports a false position or time without any warning.

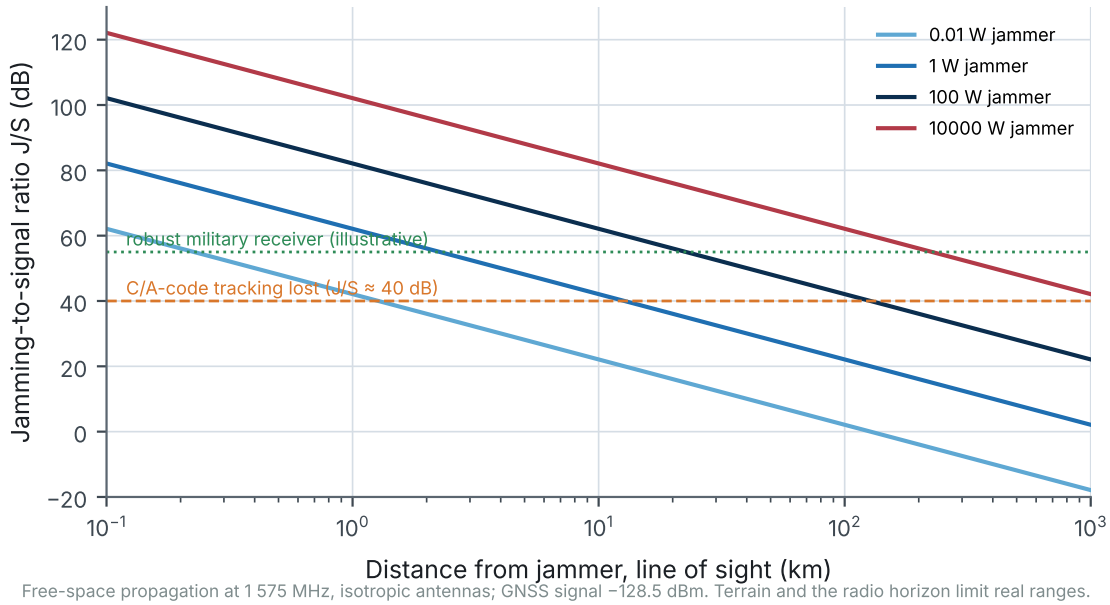


FIGURE 18.16 Jamming-to-signal ratio versus distance from jammers of different power. Even very low-power jammers disrupt receivers at kilometres; high-power military jammers reach the radio horizon.

Since 2017, interference has grown into a serious international problem. Ships in the Black Sea reported spoofed positions placing them at inland airports; aircraft over the Middle East, the Baltic region and elsewhere routinely encounter jamming and spoofing linked to military conflicts. In Lithuania alone, more than 1 000 aircraft per month reported interference in the first half of 2025. The number of spoofing incidents in the region rose to several thousand per month in mid-2025, with transmitters located in the Kaliningrad region. Aviation authorities (ICAO, EASA, the FAA) have issued safety bulletins, and airlines have had to return to traditional radio navigation aids.

Countermeasures operate at several levels:

- **Robust receivers**: multi-frequency, multi-constellation tracking; interference detection and filtering; controlled-reception-pattern antennas (CRPAs) that steer nulls towards jammers; integration with inertial sensors that bridge outages.
- **Signal authentication**: Galileo's **Open Service Navigation Message Authentication (OSNMA)**, operational since 2025, digitally signs the navigation data so that receivers can detect forged messages. GPS plans similar protection (CHIMERA), and encrypted military signals (GPS M-code, Galileo PRS) resist spoofing.
- **Complementary PNT**: terrestrial systems (eLoran, mobile-network timing, fibre time distribution), signals from LEO satellites, and new technologies such as quantum inertial sensors and magnetic-anomaly navigation. The aim is that critical infrastructure does not depend on GNSS alone. US Executive Order 13905 (2020) requires critical-infrastructure operators to manage their PNT risks.

18.8 Future Systems

18.8.1 Modernisation

All the global systems are modernising. The final GPS III satellite (SV10) was launched in April 2026. The next generation, **GPS IIIF**, from about 2027, will add a regional high-power military signal, laser retroreflectors for precise orbit determination, and search-and-rescue payloads. Europe's **second-generation Galileo** satellites, built by Airbus and Thales Alenia Space, introduce electric propulsion, fully digital reconfigurable payloads, improved clocks and inter-satellite links, with first launches expected in 2026–27. China plans a next-generation BeiDou around 2035 with LEO augmentation, and GLONASS is moving to CDMA signals with GLONASS-K2. Japan is expanding QZSS to seven and later eleven satellites. India is replacing NavIC satellites with the NVS series, which carry indigenous atomic clocks and an L1 signal for wider receiver compatibility.

18.8.2 LEO PNT

Satellites in low Earth orbit are about 20 times closer than MEO navigation satellites. So their signals arrive about 30 dB stronger, which makes them much harder to jam. Their faster motion across the sky also allows rapid convergence of precise solutions. **Iridium**'s Satellite Time and Location service already provides timing and positioning that works indoors. Start-ups such as **Xona Space Systems** (which launched its first production-class Pulsar satellite in 2025) are building dedicated LEO PNT constellations. ESA's **LEO-PNT** in-orbit demonstrators (Celeste) are testing the concept. Researchers have also shown that even signals from broadband constellations such as Starlink can be used opportunistically for navigation. LEO PNT is expected to complement rather than replace MEO GNSS.

18.8.3 Navigation Beyond the Earth

GNSS signals are increasingly used above the constellations. In geostationary and high orbits, receivers use the weak signals that spill past the Earth's limb and the antenna side lobes — the **space service volume**. The **Lunar GNSS Receiver Experiment (LuGRE)**, an Italian–NASA receiver, flew on Firefly's Blue Ghost lander. In March 2025 it became the first to track GPS and Galileo signals on the Moon's surface and compute a navigation fix, some 380 000 km from Earth (Figure 18.17). For the coming era of lunar exploration, NASA, ESA (Moonlight), JAXA and others are developing **LunaNet**, a framework for lunar communications and navigation with dedicated lunar navigation satellites and a lunar time standard.

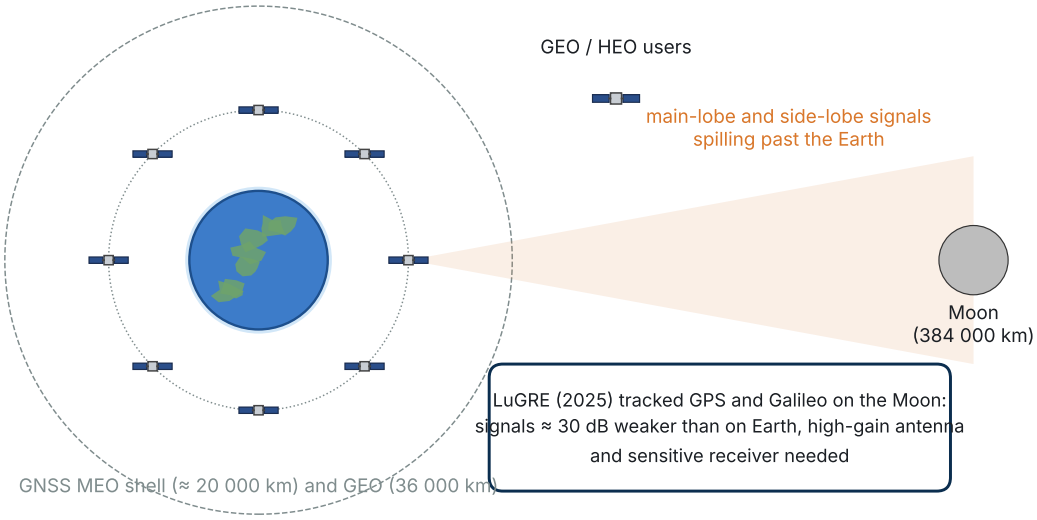


FIGURE 18.17 GNSS signals beyond the constellations. Users in high orbits and at the Moon receive signals that spill past the Earth's limb.



FIGURE 18.18 Firefly's Blue Ghost Mission 1 lander was launched on 15 January 2025 on a Falcon 9 rocket. It carried LuGRE, which made the first GNSS navigation fix on the Moon in March 2025. Image: NASA/KSC (NASA/Kim Shiflett).



FIGURE 18.19 The Gravity Probe B spacecraft, launched in 2004 on a Delta II, confirmed predictions of general relativity — the same physics that every GNSS satellite clock must account for. Image: NASA/MSFC.

EXAMPLE: WHAT ACCURACY DO GNSS CLOCKS AND ORBITS NEED?

1. *Clock stability.* Suppose the satellite clock contribution to the range error must stay below 0.3 m (1 ns) for 12 hours between uploads. Then the clock must keep time to 1 ns in 43 200 s. This is a fractional frequency stability of about $1 \times 10^{-9} / 43\,200 \approx 2 \times 10^{-14}$. Only hydrogen masers and the best rubidium clocks achieve this; the ground segment also estimates and broadcasts each clock's drift.
2. *Orbit accuracy.* An orbit error mainly affects the range along the line of sight. To contribute less than 0.3 m, the broadcast orbit must be known to about 0.3 m radially — even though the satellite moves at 3.9 km/s.
3. *Position accuracy.* With UERE = 1.0 m (dual-frequency) and PDOP = 1.5 (multi-GNSS), the 1-σ position error is about 1.5 m. With RTK (UERE ≈ 1 cm for carrier phase after ambiguity resolution), it falls to about 1.5 cm.
4. *Timing.* A static receiver at a known position needs only one satellite to solve for time; averaging many satellites gives timing accuracy of 10–20 ns relative to UTC — sufficient for 5G networks.

AI IN SATELLITE NAVIGATION

Machine learning helps receivers detect and classify interference and spoofing, mitigate multipath in cities by learning which signals are reflected, fuse GNSS with cameras, lidar and inertial sensors in autonomous vehicles, and forecast ionospheric conditions. On the ground, AI models improve orbit and clock predictions and monitor the health of the constellation. Chapters 47 and 49 discuss these applications.

KEY TAKEAWAYS

- GNSS provides positioning, navigation and timing worldwide; about seven billion receivers depend on it, and so do telecom, power and financial networks.
- Receivers measure pseudoranges to at least four satellites and solve for position and receiver clock offset; carrier phase allows centimetre precision.
- Signals are weaker than thermal noise but are recovered through spread-spectrum correlation; codes such as the GPS C/A Gold code allow all satellites to share frequencies.
- GPS, GLONASS, Galileo and BeiDou each operate about 24–30 MEO satellites with atomic clocks; QZSS and NavIC serve regions. Relativity changes GPS clock rates by +38.5 µs per day.
- Accuracy depends on the error budget (ionosphere, clocks, orbits, troposphere, multipath) and satellite geometry (DOP); dual-frequency, multi-GNSS receivers achieve 1–2 m.
- SBAS, GBAS, RTK and PPP improve accuracy to centimetres and provide integrity for aviation and other safety-critical uses.
- Jamming and spoofing have become widespread; authentication (Galileo OSNMA), robust receivers and complementary PNT (including LEO PNT) are the responses.
- GNSS is now used at the Moon: LuGRE made the first lunar GNSS fix in 2025.

REVIEW QUESTIONS

1. Why does a GNSS receiver need signals from at least four satellites? What are the four unknowns?
2. The GPS C/A code has a chipping rate of 1.023 Mchip/s and carries data at 50 bit/s. What processing gain does despreading provide, in decibels?
3. GPS clocks gain about 38.5 µs per day relative to the ground because of relativity. What ranging error would accumulate in one day if this were not corrected?

4. A receiver sees satellites with a PDOP of 1.8, and the user equivalent range error is 1.2 m. Estimate the position error.
5. List the main sources in the GNSS error budget and explain how dual-frequency receivers remove the largest of them.
6. Compare SBAS, RTK and PPP in accuracy, coverage and what they need from the user.
7. How do jamming and spoofing work, and how does Galileo OSNMA help against spoofing?
8. Why are low-Earth-orbit PNT systems being developed, and what did LuGRE demonstrate at the Moon?